

Episodic Factor Pricing^{*}

Sophia Zhengzi Li[†], Peixuan Yuan[‡], and Guofu Zhou[§]

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ABSTRACT

Conventional factor models assume that risk factors are priced consistently over time. We challenge this view by showing that factor pricing power switches between active and inactive regimes. We develop a real-time method to identify *ex ante* whether a factor or factor model is “pricing on” or “pricing off.” We find that factor premia are concentrated almost entirely in active states and are largely absent otherwise. For instance, the Fama-French models are active in only about half of the sample. Conditioning on these states delivers economically and statistically significant out-of-sample performance gains. These findings establish that factor pricing is episodic rather than persistent.

JEL classification: G11, G12, G14

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[†]Rutgers Business School; E-mail: zhengzi.li@business.rutgers.edu.

[‡]School of Business, Hong Kong Baptist University; E-mail: peixuan@hkbu.edu.hk

[§]Olin Business School, Washington University in St. Louis; E-mail: zhou@wustl.edu.

I. Introduction

Empirical asset pricing typically assumes that pre-specified factors have the same risk premia over time (see, e.g., Sharpe, 1964; Lintner, 1965; Black, 1972; Fama and French, 1993; Lustig, Roussanov, and Verdelhan, 2011; Fama and French, 2015; Hou, Xue, and Zhang, 2015; Stambaugh and Yuan, 2017; Daniel, Hirshleifer, and Sun, 2020). To better capture the empirical performance of the factors, a large literature relaxes this assumption by allowing factor exposures or risk premia to vary with firm characteristics or economic conditions (e.g., Ferson and Harvey, 1991; Lettau and Ludvigson, 2001; Gomes, Kogan, and Zhang, 2003; Avramov and Chordia, 2006a; Kelly, Pruitt, and Su, 2019).

In this paper, we investigate time variation in factor pricing power. Rather than modeling continuously time-varying exposures or prices of risk, we ask whether pricing power is episodic and can be identified in real time using a rich information set. Instead of forecasting factor returns, we model the state of pricing power directly and allow it to switch between two regimes, which we label pricing-on and pricing-off, or simply “on” or “off.”

At each time t , we determine whether a factor or factor model is pricing-on or pricing-off by assessing whether it improves the explanation of the cross-section of stock returns relative to a benchmark without the factor.¹ The intuition is straightforward: if a factor’s underlying characteristic fails to reduce cross-sectional forecast errors, the factor is deemed inactive for pricing and investment purposes. We quantify the gain using the difference in cross-sectional mean squared forecast errors with and without the factor. This forecast-error difference (SED) is predicted in real time, ex ante. If the predicted $SED > 0$, the factor is classified as “on”; otherwise, it is “off.”

We identify these states in real time using a fully out-of-sample procedure. Following Fama and MacBeth (1973), we estimate a sequence of cross-sectional univariate regressions in which firm-level factor characteristics are used to predict next-month stock

¹In standard time-series asset pricing, factors are motivated by average slopes from such cross-sectional regressions linking returns to characteristics: for example, the three-factor model of Fama and French (1993) incorporates size and book-to-market, while the five-factor model of Fama and French (2015) adds profitability and investment to capture return patterns unexplained by earlier models (Sharpe, 1964; Lintner, 1965; Titman, Wei, and Xie, 2004; Novy-Marx, 2013).

returns. We then forecast the next-month SED using only information available at time t via a LASSO regression. The model selects from a broad set of predictors, including macroeconomic variables (Welch and Goyal, 2008), investor sentiment measures (Baker and Wurgler, 2006), and lagged SED at multiple horizons. These predictors capture variation in macroeconomic conditions, investor sentiment, and innovations in factor pricing power.

Our approach is inspired by Farmer, Schmidt, and Timmermann (2023). However, whereas their analysis focuses on the time-series predictability of the aggregate market return, we examine the cross-sectional pricing power of factors. Moreover, unlike their reliance on lagged predictors alone, we incorporate both market-wide and factor-specific information and employ regularization to mitigate overfitting.

We begin by examining the pricing power of the Fama and French (1993) three-factor (FF3) and Fama and French (2015) five-factor (FF5) models across active and inactive regimes. Prior studies typically treat these models as time invariant, implicitly assuming that their pricing power is constant over time. We estimate the models recursively to generate out-of-sample return forecasts based on model-implied characteristics. Using the resulting history of SED, we identify pricing-on and pricing-off states for each factor model in real time.

Surprisingly, we find that the FF3 model is in a pricing-on state only about 50% of the time (295 out of 588 months). The FF5 model performs slightly better but is still active only 53% of the time. While these frequencies are lower than might be expected for such standard models, likely reflecting the difficulty of real-time prediction, our classification effectively isolates periods of genuine pricing efficacy. First, the pricing power of both models is concentrated almost entirely in pricing-on periods. For example, the FF3 model generates a significantly positive out-of-sample cross-sectional R^2 during pricing-on months, but a negative R^2 during pricing-off periods. Second, economically, a long-short portfolio constructed from FF3-based return forecasts earns an annualized Sharpe ratio of 0.95 when pricing is on, but is unprofitable when pricing is off, yielding a Sharpe ratio of -0.28 . This disparity translates into massive economic gains: for an

initial \$1 investment, a dynamic timing strategy that switches to the market during inactive periods accumulates a terminal wealth of \$110.06, an order of magnitude larger than the \$7.31 generated by the static strategy. Overall, the FF3 model explains cross-sectional differences in average returns and delivers meaningful risk premia primarily during pricing-on states, highlighting the economic importance of distinguishing between active and inactive regimes.

Because pricing states need not align across factors, we next examine the pricing power of individual Fama-French factors within their own pricing-on and pricing-off periods.² The size factor earns an average return of 0.73% ($t = 3.66$) in pricing-on months, but -0.21% ($t = -1.31$) in pricing-off months. This contrast helps reconcile the size factor’s weak unconditional performance. While prior work documents a weak and time-varying size premium (Israel and Moskowitz, 2013), subsequent studies show that size remains economically meaningful when appropriately conditioned, either by controlling for firm quality or accounting for cash-flow shocks (Asness, Frazzini, Israel, Moskowitz, and Pedersen, 2018; Hou and van Dijk, 2018). Our results complement this literature by showing that size premia are concentrated in periods when pricing power is active.

The value factor exhibits a similar pattern, earning 0.60% ($t = 3.94$) during pricing-on periods and -0.29% ($t = -1.10$) during pricing-off periods, consistent with evidence of a declining or unstable value premium (Fama and French, 2021; Smith and Timmermann, 2022). Similar state dependence is observed for the five-factor model (FF5). The profitability and investment factors earn economically and statistically significant returns during pricing-on periods, with average returns of 0.51% ($t = 5.18$) and 0.38% ($t = 3.13$), respectively. Moreover, the risk–return trade-off—measured as next month’s realized factor return relative to the realized variance of daily returns in the previous month, is materially stronger during pricing-on periods for all FF5 factors and weak during pricing-off periods.

²We analyze all Fama-French factors except the market factor, as our identification approach focuses on cross-sectional predictability from firm characteristics and associated factor pricing power. Predictability of the aggregate market return is the focus of the market-timing literature, which studies a distinct object; see, among others, Welch and Goyal (2008), Rapach, Strauss, and Zhou (2010), Henkel, Martin, and Nardari (2011), Pettenuzzo, Timmermann, and Valkanov (2014), and the survey by Rapach and Zhou (2022).

Our approach generalizes to alternative pricing models and their constituent factors. Specifically, the episodic pattern extends to the six-factor model (FFC6), which augments the five-factor model with the momentum (UMD) factor of Carhart (1997), as well as the q-factor model of Hou, Xue, and Zhang (2015), which relies on profitability (ROE) and investment (IA) factors. It also extends to the nine-factor model studied by Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024).³ We find that similar results also hold for both the aggregate models and the associated factors. Importantly, these findings are not driven by variation in market volatility. When the sample is partitioned into low- and high-volatility terciles, pricing-on months continue to deliver high Sharpe ratios in both regimes, whereas pricing-off months do not. Appraisal ratios remain economically large across volatility states, indicating that pricing states capture information beyond market volatility.

To understand the economic mechanism underlying these gains, we further examine the asset pricing implications of episodic factor pricing. The central mechanism is that pricing-on states correspond to periods in which factor-implied firm characteristics meaningfully differentiate expected returns across assets, indicating higher prices of risk and stronger compensation for bearing factor risk, whereas this mapping breaks down in pricing-off states. We find that the risk-return trade-off strengthens when a factor is pricing-on and weakens when it is pricing-off. Accordingly, a mean-variance investor should scale down factor exposures during pricing-off periods.⁴ Consistent with this implication, our pricing-state-managed multifactor portfolios reduce exposures when a factor is predicted to be pricing-off and outperform unmanaged benchmarks that ignore pricing states, both out of sample and net of transaction costs. For example, for the FF3 model, the Sharpe ratio increases from 0.50 to 0.88. For the FF5 model augmented with UMD, the Sharpe ratio rises from 1.06 to 1.31. Using the nine-factor set studied by Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024), the managed Sharpe ratio

³Latent factors, such as Ross (1976), Kelly, Pruitt, and Su (2019) and Lopez-Lira and Roussanov (2023), are more complex and will be a topic of future research.

⁴Our approach is related to recent work integrating machine learning techniques into mean-variance portfolio construction (see, e.g., Wang, Gao, Harvey, Liu, and Tao, 2026), although our focus is on state-dependent factor pricing rather than direct return prediction.

increases from 1.18 to 1.56, and the appraisal ratio reaches 1.01.

To formalize this channel, we estimate a conditional stochastic discount factor in which each factor’s price of risk is an affine function of its pricing state. In line with the literature, the unconditional price of risk is positive for each factor. Importantly, the conditional price of risk is markedly higher when the factor is pricing-on, and these results remain robust after controlling for realized market volatility. This on–off structure provides a simple and tractable way to condition the SDF on factor-specific pricing states and meaningfully expands the span of the investment opportunity set.

An asset pricing model with a higher Sharpe ratio is unambiguously superior, as it necessarily delivers smaller aggregate pricing errors (Barillas and Shanken, 2017). The substantially higher Sharpe ratio of the stochastic discount factor (SDF) that incorporates pricing states therefore implies that permitting factors to switch between pricing-active and pricing-inactive regimes leads to a significant improvement in model performance relative to conventional constant-pricing benchmarks. Taken together, this evidence challenges the standard assumption of constant factor pricing power and highlights the empirical relevance of state-dependent factor pricing.

Our paper contributes to empirical asset pricing by proposing a novel framework in which factors can switch between pricing-active (“on”) and pricing-inactive (“off”) states.⁵ In contrast to a large class of influential asset pricing models, such as Fama and French (1993, 2015); Hou, Xue, and Zhang (2015); Stambaugh and Yuan (2017); Daniel, Hirshleifer, and Sun (2019), which implicitly assume that factors are always priced, our framework allows factors to be intermittently active. Empirically, this distinction is critical: factors frequently transition from pricing-on to pricing-off states and back again. Our results suggest that factor pricing power is inherently episodic rather than persistent and point to potentially broad applications of the framework to other asset classes, including bonds, currencies, and options.

Our paper also contributes to the literature on the predictability of factor returns. Fac-

⁵This concept is distinct from the widely used “risk-on/risk-off” framework, which refers to shifts in investor risk appetite and attention that lead to more aggressive or more defensive investment behavior (see, e.g., Chari et al., 2023).

tor investing has received substantial attention from both researchers and practitioners, and recent studies suggest that factors relevant for portfolio choice can be identified and timed based on their predictability (see, e.g., Haddad, Kozak, and Santosh, 2020; Gupta and Kelly, 2019; Ehsani and Linnainmaa, 2022; Arnott, Kalesnik, and Linnainmaa, 2023; Kelly, Malamud, and Pedersen, 2023; Chen, Liu, Wang, Wang, and Yu, 2025). More recently, Farmer, Schmidt, and Timmermann (2023) document short-lived pockets of market return predictability driven by sticky investor expectations and provide evidence of predictability for the size and value factors. Other studies relate factor performance to factor-specific or aggregate market volatility (see, e.g., Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016; Cederburg and O’Doherty, 2016; Moreira and Muir, 2017; DeMiguel, Martín-utrera, and Uppal, 2024; Barroso, Detzel, and Maio, 2025). In contrast to this literature, we develop a real-time method to identify the pricing state of individual factors and factor models using a rich information set. Our approach relies on a transparent measure of cross-sectional forecast accuracy and employs LASSO regularization to mitigate overfitting. We document that conditioning factor exposures on pricing states leads to substantial improvements in multifactor portfolio performance out of sample and net of transaction costs.

Our paper also relates to the literature on cross-sectional stock return predictability. Recent work applies machine learning methods to improve return forecasts by addressing the high dimensionality of the cross section and capturing nonlinear relationships among predictors (see, e.g., Light, Maslov, and Rytchkov, 2017; Kozak, Nagel, and Santosh, 2018, 2020; Kelly, Pruitt, and Su, 2019; Freyberger, Neuhierl, and Weber, 2020; Gu, Kelly, and Xiu, 2020; Cong, Feng, He, and He, 2025; Chen, Pelger, and Zhu, 2024). We complement this literature by showing that an ensemble framework that selectively aggregates individual characteristics based on their predictive content delivers significant economic value. In addition, we provide evidence that macroeconomic conditions and market-wide investor sentiment play a central role in shaping cross-sectional return predictability (see, e.g., Avramov and Chordia, 2006a,b; Shen, Yu, and Zhao, 2017; Smith and Timmermann, 2021; Gómez-Cram, 2022; Chen, Liu, Wang, Wang, and Yu, 2025).

The remainder of the paper is organized as follows. Section II describes the data and the methodology used to forecast the cross-section of stock returns and to identify factor pricing states. Section III presents evidence from the Fama-French factor models. Section IV extends the analysis to alternative pricing specifications. Section V examines the asset pricing implications of episodic factor pricing. Section VI concludes. The Appendix reports supplementary empirical evidence that complements the main results.

II. Data and Methodology

A. Data

The stock return data are obtained from the Center for Research in Security Prices (CRSP) monthly stock file. The analysis focuses on common stocks of companies incorporated in the United States that are traded on NYSE, AMEX, or Nasdaq, adhering to standard conventions. The characteristic data are sourced from the comprehensive online dataset provided by Chen and Zimmermann (2022).

We also collect the same nine factors as Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024). We download excess returns from the authors' websites for the market (MKT), small-minus-big (SMB), high-minus-low (HML), robust-minus-weak (RMW), conservative-minus-aggressive (CMA), and momentum (UMD) factors of Fama and French (2018); the profitability (ROE) and investment (IA) factors of Hou, Xue, and Zhang (2015); and the betting-against-beta (BAB) factor of Frazzini and Pedersen (2014).

Our sample period spans from January 1970 to December 2023. The initial in-sample period for estimating the forecasting models consists of the first five years. Accordingly, we compute and evaluate out-of-sample return forecasts from January 1975 to December 2023.

B. Cross-Sectional Return Forecasts

Factors in time-series asset pricing models are often motivated by evidence from Fama and MacBeth (1973, FM) cross-sectional regressions that average returns are related to asset characteristics.⁶ Therefore, we identify factors’ pricing power by detecting the cross-sectional return predictability of the associated firm characteristic. To construct out-of-sample one-month-ahead return forecasts for month $t+1$ based on firm characteristic j , denoted as $C_{j,i,t}$, for firm i at month t , we adopt the FM regressions also employed by Lewellen (2015). This ensures that all forecasts are based on information available at the time of forecast formation, avoiding any “look-ahead” bias. Our objective is to assess the time-varying predictability of firm characteristics in determining returns. To this end, we estimate a sequence of cross-sectional univariate regressions, where each regression relates returns to an individual characteristic. The model is given by:

$$r_{i,t} = a_{j,t} + b_{j,t}C_{j,i,t-1} + \varepsilon_{i,t}, \quad (1)$$

where $r_{i,t}$ represents the excess return of stock i at time t . Following Kelly, Pruitt, and Su (2019) and Kozak, Nagel, and Santosh (2020), we employ a cross-sectional transformation of characteristics on a period-by-period basis. Specifically, we compute the ranks of stocks for each characteristic, divide the ranks by the number of non-missing observations, and subtract 0.5. This mapping procedure ensures that the characteristics are standardized within the interval of $[-0.5, +0.5]$, emphasizing their ordering rather than their magnitude. We adopt this standardization approach due to its robustness against outliers.

To enhance the efficiency of the slope estimators, we employ market capitalization at the end of the previous month as regression weights. Following the standard approach of the FM method, we estimate Equation (1) on a period-by-period basis up until month t when making predictions for month $t+1$. The forecasted cross-sectional return for month

⁶For example, the three-factor model of Fama and French (1993) incorporates size and book-to-market ratio, characteristics shown to capture return variations missed by the CAPM (Sharpe, 1964; Lintner, 1965). Similarly, the five-factor model of Fama and French (2015) adds profitability and investment, motivated by evidence that these explain return patterns beyond the three-factor model (Titman, Wei, and Xie, 2004; Novy-Marx, 2013).

$t+1$ is given by:

$$\hat{r}_{j,i,t+1|t} = \bar{a}_{j,t} + \bar{b}_{j,t} C_{j,i,t}, \quad (2)$$

where

$$\bar{a}_{j,t} = \frac{1}{M} \sum_{m=0}^{M-1} \hat{a}_{j,t-m}, \quad \bar{b}_{j,t} = \frac{1}{M} \sum_{m=0}^{M-1} \hat{b}_{j,t-m}, \quad (3)$$

and M is the number of regressions estimated up until month t . Equation (2) represents the conventional Fama-MacBeth out-of-sample cross-sectional return forecast, as employed by Haugen and Baker (1996), Lewellen (2015), and Green, Hand, and Zhang (2017). As noted by Han, He, Rapach, and Zhou (2024), smoothing the coefficient estimates over time helps stabilize the coefficient estimates and, consequently, improves the cross-sectional return forecasts and reduces the risk of overfitting.

C. Detecting the State of Factor Pricing

In the study of time-series predictability of the aggregate market excess return, Farmer, Schmidt, and Timmermann (2023) finds that macro predictors exhibit significant short-horizon return predictability during brief periods due to the sticky expectations of investors. They propose a recursive and out-of-sample approach to identify these periods, utilizing only real-time information available prior to the period under consideration for return prediction. In this paper, we conduct a cross-sectional analysis and examine the explanatory power of firm characteristics on the cross-sectional dispersion in stock returns. We conjecture that there exist effectively binary states of factor pricing power, used as a parsimonious classification device rather than a structural restriction. We define “pricing-on” regimes as periods when the factor exhibits cross-sectional pricing power and “pricing-off” regimes as periods in which such pricing power is not statistically detectable relative to a benchmark.

Specifically, we first define a measure of the relative predictive accuracy of the firm characteristic C_j in month t as the cross-sectional mean squared forecast error difference (SED) given by:

$$SED_{j,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left[(dr_{i,t})^2 - \left(dr_{i,t} - \hat{dr}_{j,i,t} \right)^2 \right], \quad (4)$$

where $dr_{i,t}$ is the demeaned stock return, calculated as $r_{i,t} - \bar{r}_t$, where $\bar{r}_t$ is the cross-sectional average of stock returns at time t and serves as the common component for all stocks; $\widehat{dr}_{j,i,t}$ is the demeaned model forecast based on (2), calculated as $\widehat{r}_{j,i,t|t-1} - \widehat{\bar{r}}_{j,t|t-1}$, where $\widehat{\bar{r}}_{j,t|t-1}$ is the cross-sectional average of return forecasts based on characteristic C_j ; N_t represents the number of stocks in month t . The benchmark forecast simply assumes a zero value for all stocks, which corresponds to forecasting only the cross-sectional mean return after demeaning, disregarding the information contained in the firm characteristics and assuming no cross-sectional dispersion. Removing the common component is crucial to capture the relative expected returns across stocks and properly time the cross-sectional return predictability of characteristics. Forecasting the common component $\bar{r}_t$ would mask the timing of cross-sectional return predictability by conflating market-wide variation with relative cross-sectional signals.⁷

Therefore, $SED_t > 0$ indicates that the predictive regression outperforms the benchmark forecast, as it delivers a smaller squared forecast error. To identify such periods when predicting the market premium, Farmer, Schmidt, and Timmermann (2023) project $SED_{j,t}$ onto a constant and a time trend. However, Cakici, Fieberg, Neumaier, Poddig, and Zaremba (2024) revisit the implementation of this forecasting specification and find that its reported out-of-sample performance weakens under correction. Building on the insight in Farmer, Schmidt, and Timmermann (2023) that predictability is episodic, we adopt a different forecasting strategy that incorporates a richer set of signals beyond a time trend, drawing on variables that prior research has documented as relevant for return predictability in the cross section:

$$SED_{j,t} = \gamma_{0,j,t} + \sum_z \gamma_{z,j,t} Z_{z,t-1} + \epsilon_{j,t}. \quad (5)$$

To stabilize the predictive regression, we first normalize $SED_{j,t}$ by the previous market

⁷As discussed in Section IV.B, the common component is critical for timing the betting-against-beta (BAB) factor of Frazzini and Pedersen (2014). To ensure market neutrality, BAB is constructed with a net-long position in the market return. Thus, its positive market exposure makes its performance resemble a market timing strategy (see, e.g., Liu, Stambaugh, and Yuan, 2018; Han, 2022). Cederburg and O'Doherty (2016) also attributes part of BAB's performance to market timing.

volatility: $SED_{j,t}/\sigma_{M,t-1}$.⁸ We consider three sets of state predictors, Z_s . First, we use 16 macroeconomic variables from Welch and Goyal (2008) to capture the state of the aggregate economy: *dividend-price ratio, dividend yield, book-to-market ratio, earnings-price ratio, Treasury bills, default yield spread, long-term rate of return, net equity expansion, stock variance, stock returns, long-term yield, inflation, default return spread, term spread, dividend payout ratio, and corporate bond return*. Second, we include five market-wide sentiment measures from Baker and Wurgler (2006): *dividend premium, first-day returns of IPOs, number of IPOs, closed-end fund discount, and share turnover*.⁹ Third, to capture trends in SED , we add the average of past SED over seven horizons (1, 3, 6, 9, 12, 18, and 24 months). In total, we use 28 variables to identify pricing-on and pricing-off periods for factor j .

To mitigate overfitting, we use LASSO, a penalized regression that induces sparsity in coefficient estimates via an L_1 penalty. Following Dong, Li, Rapach, and Zhou (2022) and Li, Yuan, and Zhou (2025), we estimate the LASSO model in an expanding window and choose the regularization parameter using the Akaike Information Criterion (AIC), which trades off in-sample fit against model complexity. For out-of-sample prediction, we estimate the predictive model (5) using only information available at time t , and predict next-month SED . We define pricing-on states as periods in which $\widehat{SED}_{j,t+1} > 0$, a threshold with a natural economic interpretation as outperformance relative to the benchmark forecast. Specifically, a pricing-on state is identified when the forecast based on C_j is considered more accurate than the benchmark, as it leads to a lower expected cross-sectional mean squared forecast error.

⁸Our results are robust to leaving SED unnormalized.

⁹We lag the constituent sentiment variables to avoid potential look-ahead bias, closely following the procedure Baker and Wurgler (2006) used to construct their sentiment index, as illustrated in the dataset obtained from their website (<https://pages.stern.nyu.edu/~jwurgler/>).

III. Evidence from Fama-French Factor Models

A. The Pricing Power of Fama and French Factor Models

We begin by assessing the out-of-sample power of two widely used factor models in pricing the cross section of stock returns: the three-factor (FF3) and five-factor (FF5) models of Fama and French (1993) and Fama and French (2015), respectively. We study the cross-sectional factor pricing model, following Fama and French (2020), who show that cross-sectional regressions are a form of factor model and provide a better description of average returns.¹⁰ FF3 includes the market, size, and value factors. FF5 augments FF3 with profitability and investment to better capture return patterns. Prior studies often treat these models as time-invariant, implicitly assuming constant pricing performance over time.¹¹

To evaluate each model's overall pricing power, we include all characteristics C_k associated with the model in the cross-sectional regressions in Equation (1):

$$r_{i,t} = a_{model,t} + \sum_{k \in model} b_{k,t} C_{k,i,t-1} + \varepsilon_{i,t}. \quad (6)$$

Specifically, for FF3, we use firms' CAPM beta, market capitalization, and book-to-market ratio. For FF5, we additionally include operating profitability and the investment-to-assets ratio. As described in Section II, we estimate the models recursively and generate out-of-sample return forecasts. Using the historical estimates of SED , we identify pricing-on and pricing-off states for each factor model in real time.

We assess the model's pricing power in the cross-section using several performance metrics. First, we compute the out-of-sample R_{OS}^2 statistic. In the cross-section, interest centers on the proportional reduction in the mean squared forecast error (MSFE) of a

¹⁰A cross-sectional factor pricing model is a monthly cross-section model in which the factors are the slope estimates from cross-sectional regressions and the factor loadings are prespecified firm characteristics. Stacking these regressions across months yields a model with time-varying factor loadings that can be used for asset pricing applications.

¹¹Recent work also emphasizes that statistical rejection of a factor model does not necessarily imply economic irrelevance and proposes economically motivated thresholds for evaluating model performance (see, e.g., Harvey, Sancetta, and Zhao, 2026).

characteristics-based forecast relative to a benchmark that predicts zero demeaned returns for all stocks. We define the monthly cross-sectional out-of-sample $R_{OS,t}^2$ for month t as:

$$R_{OS,t}^2 = 1 - \frac{\sum_{i=1}^{n_t} (dr_{i,t} - \hat{dr}_{i,t})^2}{\sum_{i=1}^{n_t} (dr_{i,t})^2}. \quad (7)$$

Equation (7) measures the proportional reduction in MSFE delivered by the competing forecast, and higher values indicate greater accuracy. Averaging over time, in the spirit of Fama-MacBeth, yields the overall cross-sectional out-of-sample R_{OS}^2 as:

$$R_{OS}^2 = \frac{1}{T} \sum_{t=1}^T R_{OS,t}^2, \quad (8)$$

where T represents the total number of months in the out-of-sample period. A positive (negative) R_{OS}^2 indicates that the competing (benchmark) forecast has, on average, a lower MSFE over the out-of-sample period.

Second, we evaluate cross-sectional bias. Following Lewellen (2015), we relate realized to predicted cross-sectional returns via:

$$r_{i,t} = \alpha_t + \delta_t \hat{r}_{i,t|t-1} + v_{i,t}, \quad (9)$$

for $1, \dots, n_t$ and $t = 1, \dots, T$, which is a cross-sectional analogue of Mincer and Zarnowitz (1969). If $\delta > 0$, forecasted returns covary positively with realized returns on average, a necessary condition for informativeness. The slope δ measures the bias in the month- t cross-sectional forecast: $\delta = 1$ implies unbiased forecasts; $\delta < 1$ ($\delta > 1$) indicates that forecasts overstate (understate) cross-sectional dispersion in expected returns. We also report the time-series average of the monthly R^2 from Equation (9):

$$R_{MZ}^2 = \frac{1}{T} \sum_{t=1}^T R_{MZ,t}^2. \quad (10)$$

Finally, we assess economic value by forming spread portfolios that go long the top-decile stocks with the highest model-predicted returns and short the bottom-decile stocks

with the lowest model-predicted returns. We report value-weighted average returns and annualized Sharpe ratios for these long-short portfolios. The average returns, thus, capture the model-implied risk premia. If the model can describe differences in average returns across assets, it should yield economically and statistically significant risk premia.

Table I
The Out-of-sample Pricing Performance

This table evaluates the out-of-sample pricing performance of the three-factor (FF3) and five-factor (FF5) models of Fama and French (1993) and Fama and French (2015), respectively. To gauge each model’s overall pricing ability, we include the characteristics associated with the model in the cross-sectional regressions in Equation (6). We estimate each model recursively and produce out-of-sample return forecasts. Using historical estimates of the cross-sectional mean squared forecast error difference, SED , we identify pricing-on and pricing-off states for each factor model in real time. The table reports the number of months in each state (N), the out-of-sample R^2_{OS} defined in Equation (8), and the slope δ and R^2_{MZ} from Equations (9) and (10). It also reports the average returns, $Mean_{LS}$, and annualized Sharpe ratios, SR_{LS} , of spread portfolios that go long the top-decile stocks with the highest model-predicted returns and short the bottom-decile stocks with the lowest model-predicted returns. The last column reports the final cumulative wealth for static and timing strategies, assuming an initial investment of \$1. The static strategy continuously holds the spread portfolio. The timing strategy adopts a dynamic approach: it holds the spread portfolio during “pricing-on” periods and switches to the market portfolio during “pricing-off” periods. To maintain consistent risk level, the market investment during off-periods is scaled *ex ante* so that its volatility matches that of the spread portfolio, estimated over a 60-month rolling window. The final wealth for the timing strategy is reported in the rows labeled “Pricing-on”. Panels A and B report performance metrics for FF3 and FF5, respectively. The sample spans January 1975 to December 2023.

	N	R^2_{OS}	$\hat{\delta}$	R^2_{MZ}	$Mean_{LS}$	SR_{LS}	Wealth
<i>Panel A: FF3</i>							
All	588	0.00 (0.02)	0.42 (1.27)	0.71	0.43 (2.25)	0.34	7.31
Pricing-on	295	0.06 (2.83)	1.59 (3.93)	0.82	1.21 (4.12)	0.95	110.06
Pricing-off	293	-0.06 (-2.85)	-0.76 (-1.55)	0.59	-0.35 (-1.46)	-0.28	
<i>Panel B: FF5</i>							
All	588	0.02 (1.22)	0.79 (3.81)	0.63	0.63 (3.09)	0.50	23.28
Pricing-on	309	0.10 (3.81)	1.63 (5.56)	0.75	1.36 (4.87)	1.06	229.02
Pricing-off	279	-0.07 (-2.96)	-0.14 (-0.54)	0.50	-0.18 (-0.68)	-0.15	

Table I presents the pricing performance of the FF3 and FF5 models, distinguishing between pricing-on and pricing-off states. Panel A reports the results for the FF3 model. Over the full sample, FF3 exhibits weak pricing power with an insignificant out-of-sample

R_{OS}^2 . The slope coefficient $\hat{\delta}$ is 0.42, indicating that the model forecasts are substantially biased. Moreover, the FF3-based long-short portfolio generates a statistically significant average return of 0.43% per month ($t = 2.25$) with an annualized Sharpe ratio of 0.34. The static strategy, which continuously holds the FF3-based long-short portfolio, generates an average return of 0.43% ($t = 2.25$) and grows an initial \$1 investment to \$7.31 over the sample period.

Conditioning on the state reveals a sharp dichotomy. During pricing-on periods ($N = 295$), FF3 delivers a statistically significant R_{OS}^2 of 0.06% ($t = 2.83$) with a slope $\hat{\delta}$ of 1.59, indicating a strong relationship between predicted and realized returns.¹² Moreover, the model-based spread portfolio performs substantially better in these states: the average return jumps to 1.21% ($t = 4.12$) with a Sharpe ratio of 0.95. In contrast, during pricing-off periods, the model yields a negative R_{OS}^2 of -0.06% and an insignificant negative return, confirming the value of avoiding these states.

To further gauge the economic value of pricing state identification, we implement a dynamic timing strategy. This strategy invests in the model-implied spread portfolio during pricing-on months and switches to a scaled market portfolio during pricing-off months. To maintain consistent risk level, the weight on the market portfolio, c_t , is determined *ex ante* such that the volatility of the market position ($c_t \cdot \text{Market}_t$) equals the volatility of the spread portfolio, calculated using a 60-month rolling window.

The last column of Panel A shows that the timing strategy yields a final cumulative wealth of \$110.06, dramatically outperforming the static strategy. This improvement is visualized in Panel A of Figure 1, where the timing strategy achieves superior long-term capital appreciation. By successfully identifying pricing-off states and switching to a market position, the timing strategy avoids the stagnation of the static factor model while capturing the equity premium during periods when the model fails to price assets cross-sectionally. Notably, the timing strategy also outperforms the stand-alone volatility-

¹²Unlike studies that report panel R_{OS}^2 to aggregate prediction errors across firms and time (e.g., Gu, Kelly, and Xiu, 2020), we focus on the cross-sectional R_{OS}^2 . This metric effectively isolates the model's ability to explain cross-sectional return dispersion. For instance, Han et al. (2024) demonstrate that a random forest model achieving a panel R_{OS}^2 of 0.33% in Gu et al. (2020) yields a cross-sectional R_{OS}^2 of only 0.01%. Viewed through this lens, a cross-sectional R_{OS}^2 of 0.06% is economically meaningful and reflects a nontrivial improvement in explaining cross-sectional return dispersion.

scaled market portfolio.

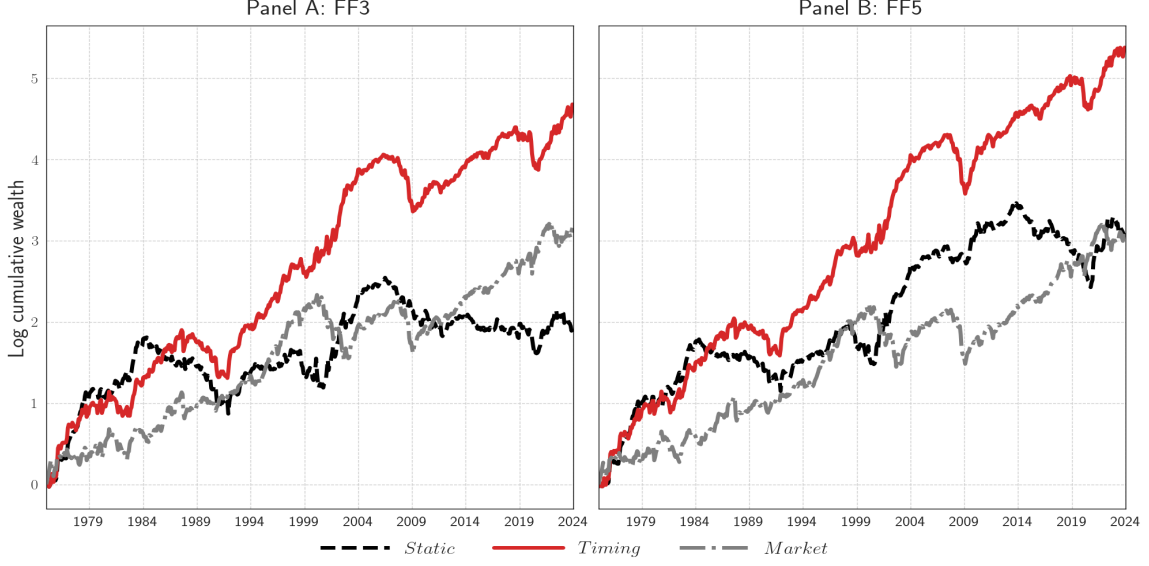


Figure 1. Cumulative Wealth of Static and Timing Strategies. This figure plots the log cumulative wealth of static and timing strategies based on the Fama-French three-factor (FF3, Panel A) and five-factor (FF5, Panel B) models. The sample period spans from January 1975 to December 2023 with an initial investment of \$1. The *Static* strategy continuously holds the long-short spread portfolio constructed from the respective model’s predicted returns. The *Timing* strategy holds the spread portfolio during identified “pricing-on” months and switches to a scaled market portfolio during “pricing-off” months. The *Market* represents a volatility-scaled market portfolio. For both the timing strategy’s off-period position and the market benchmark, the market portfolio weight is determined *ex ante* such that its volatility matches that of the spread portfolio (estimated over a 60-month rolling window).

Panel B of Table I presents the results for the FF5 model. Consistent with prior literature, FF5 improves upon FF3, delivering a full-sample R_{OS}^2 of 0.02% and a slope $\hat{\delta}$ of 0.79. The static FF5-based portfolio earns a mean return of 0.63% ($t = 3.09$) and grows the initial \$1 investment to \$23.28. However, the performance is again highly state-dependent. During pricing-on months ($N = 309$), FF5 achieves a robust R_{OS}^2 of 0.10% ($t = 3.81$) and a slope $\hat{\delta}$ of 1.63. The economic gains in this state are remarkable: the portfolio earns 1.36% per month ($t = 4.87$) with a Sharpe ratio of 1.06. Consequently, the timing strategy based on FF5 generates a final wealth of \$229.02, far outperforming the static strategy.

Panel B of Figure 1 displays the cumulative wealth of FF5-based trading strategies. As shown in the figure, the FF5-based timing strategy similarly dominates the alternatives. While the static FF5 portfolio performs relatively well, the timing strategy leverages the pricing-on states to generate significant alpha and preserves value during pricing-off

states by pivoting to the scaled market portfolio. Consequently, the cumulative wealth of the timing strategy ends substantially higher than both the static strategy and the benchmark market portfolio, further validating the economic significance of real-time pricing state identification.

Therefore, both models demonstrate significant predictive ability and economic value primarily during pricing-on periods. The stark contrast in performance metrics between the two states validates the effectiveness of our methodology in identifying periods where factor models successfully describe the cross-section of returns.

B. Model State and Economic Conditions

In the previous section, we established that the standard Fama and French factor models exhibit episodic pricing power, successfully explaining the cross-section of returns only during distinct “pricing-on” regimes. To better understand the economic mechanisms driving these transitions, we now examine the feature importance of the state predictors used in our real-time identification. By opening the “black box” of our forecasting model, we investigate which macroeconomic variables, sentiment proxies, and historical performance metrics dictate whether a factor model is active or inactive.

To quantify the relevance of each signal, we examine the coefficients from the LASSO regressions used to predict the *SED*. Since the LASSO penalization shrinks uninformative coefficients to zero, the magnitude of a surviving coefficient reflects its predictive importance.¹³ For a given predictor z , we calculate its absolute coefficient at each time t , denoted as $|\hat{\beta}_{z,t}|$. We then compute its time-series average over the full sample, $\bar{\beta}_z = \frac{1}{T} \sum_t |\hat{\beta}_{z,t}|$, and normalize it by the sum of all averaged absolute coefficients to obtain its relative contribution: $RC_z = \frac{\bar{\beta}_z}{\sum_j \bar{\beta}_j}$.

Figure 2 visualizes this feature importance for the FF3 model. Panel A ranks the predictors by their overall relative contribution (RC_z) across the entire sample. The top-ranked signals are heavily dominated by the model’s own historical performance (e.g., Past 9-month, 18-month, 12-month, and 1-month *SED*) as well as broad market condi-

¹³Note that the state predictors are standardized before each expanding-window LASSO fit.

tions, with “Stock returns” and “Number of IPOs” ranking among the most important exogenous predictors. Panel B provides a dynamic view of this feature importance over time by plotting the annual average of each predictor’s normalized absolute coefficient. The figure reveals that while the importance of certain signals fluctuates with macroeconomic cycles, the reliance on past *SED* and aggregate stock returns remains consistently strong across the decades. The feature importance for the FF5 model exhibits a very similar structure and is reported in the Figure IA.1 in the Internet Appendix for brevity.

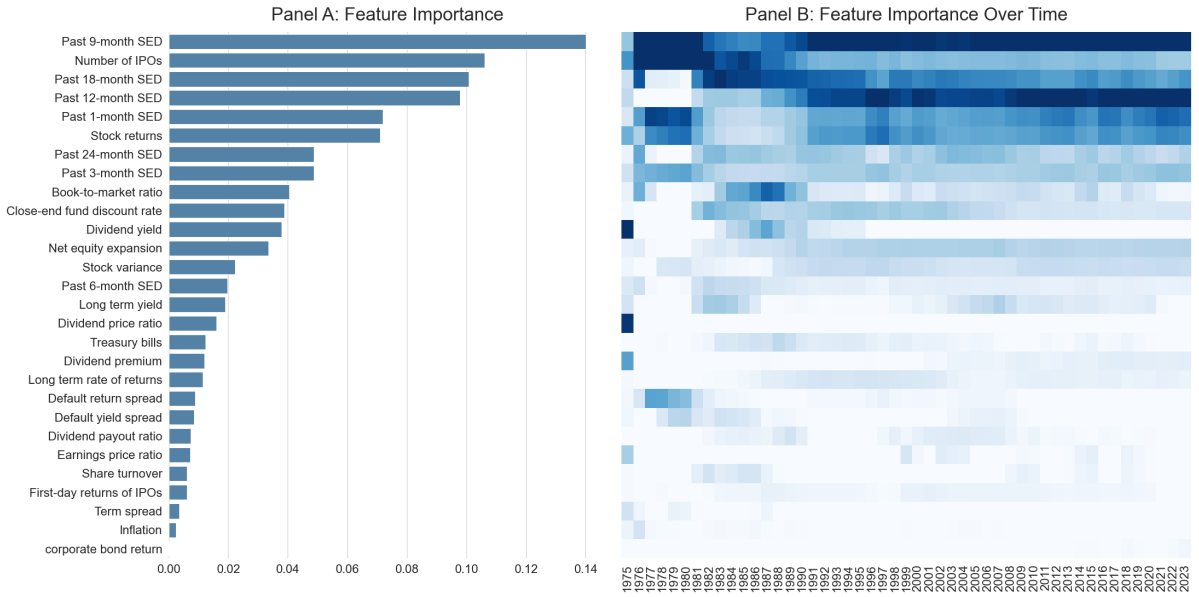


Figure 2. Feature importance for the FF3 model. This figure illustrates the relative importance of state predictors in forecasting the FF3 model’s *SED*. Panel A reports the relative contribution of each predictor, calculated as the time-series average of its absolute LASSO coefficient divided by the sum of all averaged absolute coefficients across all predictors. Panel B displays a heatmap of these relative contributions over time, calculated as the annual average of the normalized absolute coefficients. Darker colors indicate a higher relative contribution in a given year. The sample spans January 1975 to December 2023.

While Figure 2 effectively identifies which variables matter, the absolute coefficients do not provide the direction of their impact. To understand how these variables affect the model’s pricing state, we run a full-sample OLS regression predicting next month’s *SED*. As explanatory variables, we select the top 10 most important predictors identified from the Figure 2. To ensure the coefficients are directly comparable, all predictors are standardized to have a mean of zero and a standard deviation of one.

Table II reports the estimated OLS coefficients, *t*-statistics, and the R^2 of these regressions for both the FF3 and FF5 models. Focusing on Panel A (FF3), the results reveal

Table II

Determinants of Pricing State for Fama-French Models

This table examines the directional impact of the top 10 selected signals in determining the Fama-French models' expected pricing performance. We adopt an in-sample approach by estimating an OLS regression where the dependent variable is next month's volatility-normalized cross-sectional mean squared forecast error difference ($SED_{t+1}/\sigma_{M,t}$). The independent variables are the top 10 most important predictors identified via LASSO. All regressors are standardized to have a mean of zero and a standard deviation of one. The table reports estimated OLS coefficients, t -statistics, and the R^2 of the fitted regressions. We scale the raw SED by 10,000 prior to normalization, so the reported coefficients reflect changes in the volatility-adjusted pricing power (with the numerator expressed in units of percentage squared, %²). The sample spans January 1975 to December 2023.

<i>Panel A: FF3</i>				<i>Panel B: FF5</i>			
Variable	Coeff.	t -stat	R^2	Variable	Coeff.	t -stat	R^2
Past 9-month SED	-4.78	-3.12	13.15	Past 9-month SED	-5.80	-3.12	13.88
Number of IPOs	-1.00	-1.34		Past 12-month SED	8.53	3.85	
Past 18-month SED	-1.39	-0.66		Number of IPOs	-1.35	-1.42	
Past 12-month SED	8.81	4.85		Past 1-month SED	5.85	5.31	
Past 1-month SED	5.04	5.45		Stock returns	5.89	6.42	
Stock returns	5.27	6.78		Past 18-month SED	-1.52	-0.85	
Past 24-month SED	0.13	0.07		Past 3-month SED	-2.39	-1.98	
Past 3-month SED	-2.22	-2.23		Close-end fund discount rate	-0.03	-0.03	
Book-to-market ratio	-0.80	-0.92		Net equity expansion	3.79	3.53	
Close-end fund discount rate	0.80	0.90		Book-to-market ratio	-0.99	-0.96	

strong procyclicality and complex autoregressive dynamics. “Stock returns” emerges as a powerful positive predictor with a coefficient of 5.27 ($t = 6.78$). Economically, a one-standard-deviation increase in past market returns is associated with an increase of 5.27 in the model's expected volatility-normalized pricing power ($SED_{t+1}/\sigma_{M,t}$) for the subsequent month. **This confirms that characteristic-based pricing performs best following market upswings, even after accounting for the prevailing market risk.** Furthermore, the model's pricing state exhibits strong short-to-medium-term persistence: a one-standard-deviation increase in “Past 1-month SED” and “Past 12-month SED” increases the predicted volatility-normalized SED by 5.04 and 8.81, respectively. Conversely, proxies for speculative market overheating, such as the “Number of IPOs” (-1.00), negatively impact the model's volatility-adjusted efficacy.

Panel B reports the OLS regression results for the FF5 model. The directional drivers are largely consistent with those of the FF3 model. “Stock returns” (5.89) and “Past 1-month SED” (5.85) remain highly significant positive predictors, reinforcing the procycli-

cal and persistent nature of factor pricing power. One notable difference is the emergence of “Net equity expansion” as a strong positive predictor (3.79, $t = 3.53$) for FF5. This suggests that the inclusion of the investment and profitability factors makes the model particularly effective during periods when corporations are actively issuing equity.

In summary, the analysis demonstrates that factor models do not operate in a vacuum. The ability of both the FF3 and FF5 models to price the cross-section of returns is highly episodic, heavily dependent on the aggregate state of the equity market, and characterized by strong short-term persistence.

C. Pricing-on/off of Individual Fama-French Factors

In the previous section, we evaluate the models’ overall performance and find that their pricing power is concentrated in pricing-on periods. These states, however, need not align across individual factors. For example, while value premia are well documented across markets and asset classes (see, e.g., Fama and French, 2012; Asness, Moskowitz, and Pedersen, 2013), the existence of a size premium remains contested. Several studies report that the size premium has a weak and time-varying track record, whereas Asness, Frazzini, Israel, Moskowitz, and Pedersen (2018) show that size matters once one controls for firm quality, with economic significance comparable to the value effect.

To reconcile these views, we analyze the pricing power of individual Fama-French factors within their own pricing-on and pricing-off periods. We implement the real-time, factor-by-factor state classification using the procedure in Section II. Specifically, for each factor, we include its associated characteristic in the cross-sectional regressions in Equation (1). We estimate the model recursively and generate out-of-sample return forecasts. Using historical estimates of the cross-sectional mean squared forecast error difference, *SED*, we identify each factor’s pricing-on and pricing-off states in real time.

Panel A of Figure 3 shows mean returns and corresponding t -statistics for each non-market factor across pricing-on and pricing-off states. All factors earn significantly positive returns in pricing-on periods. The size factor delivers an average return of 0.73% with a t -statistic of 3.66 in pricing-on months, but -0.21% ($t = -1.31$) in pricing-off months.

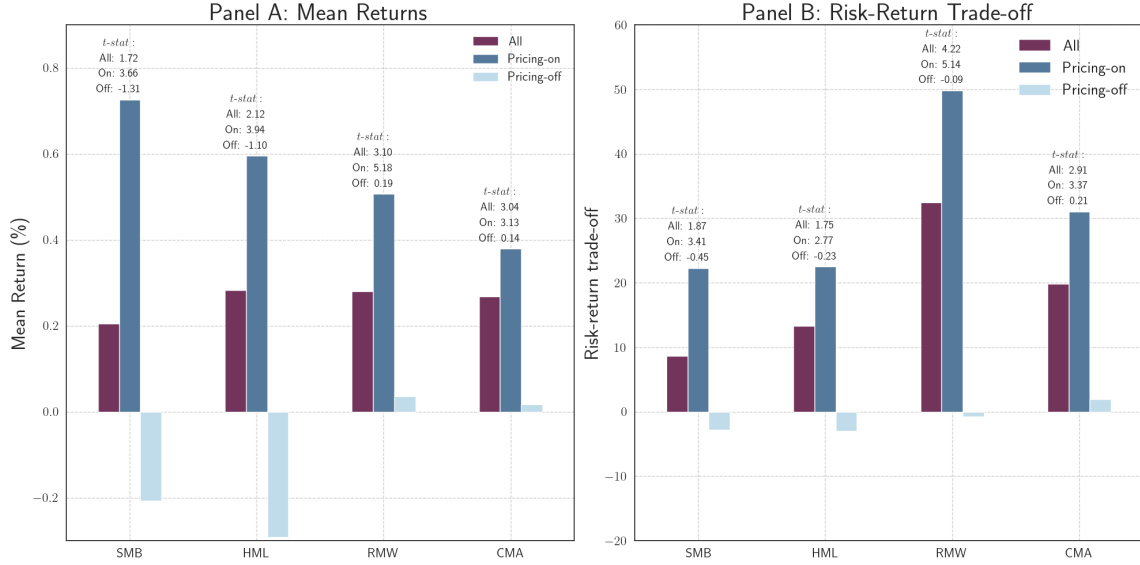


Figure 3. Pricing-on/off of Fama-French factors. This figure reports the performance of each non-market factor in the FF5 model of Fama and French (2015) across pricing-on and pricing-off periods. Panel A compares the average monthly return of each factor in the all, pricing-on, and pricing-off periods. Panel B compares the risk-return trade-off across states. We measure the trade-off as the realized factor return in month $t + 1$ divided by the realized factor variance in month t , where variance is the sample variance of daily returns within month t . The sample spans January 1975 to December 2023.

This divergent effect helps explain the size factor’s weak unconditional performance and limited pricing power in the model. Thus, our results complement Asness, Frazzini, Israel, Moskowitz, and Pedersen (2018) by showing that size matters when its pricing power is “turned on.” The value factor shows similar state dependence: 0.60% ($t = 3.94$) during pricing-on periods versus -0.29% ($t = -1.10$) during pricing-off periods. The profitability and investment factors also post strong pricing-on returns of 0.51% ($t = 5.18$) and 0.38% ($t = 3.13$), respectively.

Panel B follows Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024) to compare each factor’s risk-return trade-off across pricing-on and pricing-off states. A mean-variance investor allocates capital based on the attractiveness of this trade-off. For each factor, we measure the trade-off as the realized factor return in month $t+1$ divided by the realized factor variance in month t , where variance is the sample variance of daily returns within month t . Strikingly, for every factor the trade-off is present only in its pricing-on months and deteriorates sharply in pricing-off months. From a portfolio-choice perspective, a standard mean-variance investor should therefore time factor states: taking larger positions when the trade-off is attractive (pricing-on)

and scaling down when it is not (pricing-off).

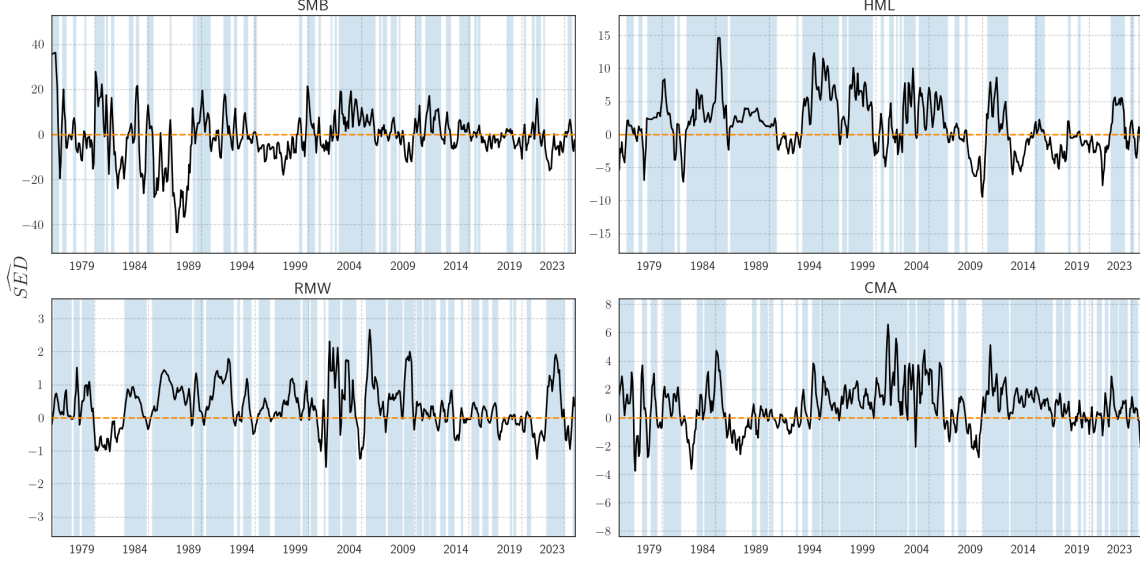


Figure 4. Pricing-on/off periods of Fama-French factors This figure plots the predicted cross-sectional mean-squared forecast error difference, $\widehat{SED}$, and the identified pricing-on periods (shaded areas) for each non-market factor in the FF5 model of Fama and French (2015). Using a rich set of market-wide and factor-specific signals, we forecast next-month $\widehat{SED}$ using Equation (5) in real time. We define pricing-on states as periods with $\widehat{SED} > 0$. The sample spans January 1975 to December 2023.

Figure 4 displays the predicted $\widehat{SED}$ and the identified pricing-on periods (shaded areas) for each factor. Pricing-on periods vary markedly across factors, with some overlap. Over January 1975 to December 2023, the fractions of months classified as pricing-on are 46%, 62%, 66%, and 64% for the size, value, profitability, and investment factors, respectively. The size factor’s weaker unconditional performance is consistent with its lower share of pricing-on months. The size factor performs particularly poorly from 1983 to 1988, when $\widehat{SED}$ is strongly negative. For the value factor, fewer pricing-on months are identified after 2008, contributing to its weaker performance since then. Some patterns of pricing-on periods, especially for size, resemble the “pockets” described in Farmer, Schmidt, and Timmermann (2023): factor pricing power is local in time, with short periods of significance interspersed with little or no pricing ability.¹⁴

¹⁴We provide a detailed discussion on the determinants of pricing states for individual factors in Section IV.B.

IV. Evidence from Alternative Pricing Models

A. The Out-of-Sample Pricing Power

In the previous section, we documented that the pricing power of the standard Fama and French (1993) and Fama and French (2015) models is highly state-dependent. A natural question arises: does this episodic behavior extend to alternative factor models? To address this, we extend our analysis to three alternative specifications: the Fama-French six-factor model (FFC6), which augments the FF5 model with the momentum factor (UMD) of Carhart (1997); the q-factor model (HXZ) of Hou, Xue, and Zhang (2015); and a combined model that aggregates factors from both specifications.

To implement the cross-sectional factor pricing model for these alternative specifications, we include the firm characteristics corresponding to each factor in the regression specified in Equation (1). For the FFC6 model, we add the momentum characteristic, defined as the cumulative return from month $t-12$ to $t-2$, to the set of FF5 characteristics. For the HXZ model, we utilize the characteristics underlying the q-factors: CAPM beta, market equity, the investment-to-assets ratio, and return on equity. Finally, the combined model encompasses the union of all unique characteristics from FFC6 and HXZ. As before, we identify pricing-on and pricing-off states in real time using the predicted *SED*.

Table III reports the out-of-sample pricing performance of these alternative models. Panel A presents the results for the FFC6 model. The inclusion of momentum significantly improves unconditional performance relative to the FF5 model reported in Table I. The all-time out-of-sample R_{OS}^2 is 0.13% ($t = 4.79$), and the model generates a robust average long-short return of 1.12% per month. However, conditioning on the pricing state reveals that this performance is unevenly distributed. In pricing-on states ($N = 409$), the R_{OS}^2 rises to 0.19% with a slope coefficient $\hat{\delta}$ of 1.23, indicating strong predictive accuracy. The spread portfolio in these states earns 1.46% per month ($t = 5.28$). In sharp contrast, during pricing-off states ($N = 179$), the predictive power vanishes: the R_{OS}^2 drops to 0.00%, and the slope coefficient $\hat{\delta}$ falls to 0.29, which is statistically indistinguishable

Table III

The Out-of-sample Pricing Performance: Alternative Models

This table evaluates the out-of-sample pricing performance of alternative models, including the FFC6 model, which augments the FF5 model with Carhart (1997)'s momentum factor, the HXZ four-factor model of Hou et al. (2015), and a comprehensive model that incorporates all factors from both FFC6 and HXZ. To gauge each model's overall pricing ability, we include the characteristics associated with the model in the cross-sectional regressions in Equation (6). We estimate each model recursively and produce out-of-sample return forecasts. Using historical estimates of the cross-sectional mean squared forecast error difference, SED , we identify pricing-on and pricing-off states for each factor model in real time. The table reports the number of months in each state (N), the out-of-sample R^2_{OS} defined in Equation (8), and the slope δ and R^2_{MZ} from Equations (9) and (10). It also reports the average returns, $Mean_{LS}$, and annualized Sharpe ratios, SR_{LS} , of spread portfolios that go long the top-decile stocks with the highest model-predicted returns and short the bottom-decile stocks with the lowest model-predicted returns. The last column reports the final cumulative wealth for static and timing strategies, assuming an initial investment of \$1. The sample spans January 1975 to December 2023.

	N	R^2_{OS}	$\hat{\delta}$	R^2_{MZ}	$Mean_{LS}$	SR_{LS}	Wealth
<i>Panel A: FFC6</i>							
All	588	0.13 (4.79)	0.94 (6.73)	0.90	1.12 (4.55)	0.64	245.21
Pricing-on	409	0.19 (5.69)	1.23 (8.25)	0.99	1.46 (5.28)	0.85	1019.85
Pricing-off	179	0.00 (-0.07)	0.29 (1.04)	0.68	0.35 (0.72)	0.20	
<i>Panel B: HXZ</i>							
	588	0.02 (1.41)	0.92 (4.82)	0.56	0.74 (4.23)	0.66	49.89
Pricing-on	313	0.08 (3.95)	1.54 (5.46)	0.64	1.19 (4.42)	1.02	165.65
Pricing-off	275	-0.05 (-2.14)	0.21 (0.97)	0.47	0.24 (1.19)	0.22	
<i>Panel C: FFC6+HXZ</i>							
	588	0.16 (5.81)	1.00 (7.31)	0.95	1.30 (5.17)	0.76	704.15
Pricing-on	422	0.20 (6.40)	1.21 (9.23)	0.96	1.52 (5.99)	0.94	1167.06
Pricing-off	166	0.06 (1.25)	0.47 (1.42)	0.93	0.72 (1.26)	0.38	

from zero. Consequently, the risk premium becomes statistically insignificant (0.35%, $t = 0.72$). The economic impact of this disparity is evident in the cumulative wealth: the static strategy yields \$245.21, whereas the timing strategy surges to \$1019.85.

Panel B reports the performance of the HXZ model. While the q-factor model is grounded in investment-based asset pricing theory, its empirical pricing power also exhibits significant time variation. Unconditionally, the model delivers a small but positive

R_{OS}^2 of 0.02%. Yet, the dichotomy between states remains strong. During pricing-on months ($N = 313$), the model achieves an R_{OS}^2 of 0.08% and a Sharpe ratio of 1.02. Conversely, in pricing-off months ($N = 275$), the R_{OS}^2 turns negative (-0.05%), and the spread portfolio return (0.24%) is statistically insignificant. This performance gap translates into distinct wealth outcomes, with the static strategy accumulating \$49.89 compared to \$165.65 for the timing strategy.

Panel C examines the comprehensive model (FFC6+HXZ). This specification yields the strongest unconditional performance among all models considered, with an all-time R_{OS}^2 of 0.16% and a Sharpe ratio of 0.76. Nevertheless, the episodic pattern persists. The model is identified as pricing-on in 422 months, during which it delivers an impressive R_{OS}^2 of 0.20% and a monthly risk premium of 1.52% ($t = 5.99$). In the remaining 166 pricing-off months, the R_{OS}^2 drops to 0.06% ($t = 1.25$), and the risk premium decreases to 0.72% ($t = 1.26$). Consequently, the timing strategy enhances terminal wealth from \$704.15 (static) to \$1167.06.

Comparison with the results in Table I highlights two key insights. First, adding the momentum factor generally increases the proportion of pricing-on months and improves unconditional metrics compared to FF3 and FF5. Second, and more importantly, the qualitative distinction between active and inactive regimes is robust across all model specifications. Regardless of the factors included, there exist distinct periods where firm characteristics fail to explain the cross-section of returns. These findings reinforce the conclusion that factor pricing is inherently episodic, and identifying these states in real time provides substantial economic value beyond the choice of the factor model itself.

B. Pricing-on/off of Representative Equity Factors

In this section, we examine the pricing-on/off properties of representative individual equity factors in Table III which are also studied in Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024). Moreira and Muir (2017) show that the market’s risk-return trade-off weakens when market volatility is high, helping to explain why a volatility-managed market portfolio outperforms. DeMiguel, Martín-utrera, and

Uppal (2024) extend this analysis beyond the market factor and show that, for all factors in their sample, the risk-return trade-off weakens with realized market volatility. This finding explains the superior performance of their conditional multifactor portfolio, which reduces factor exposures when realized market volatility is high, relative to its unconditional counterpart.

Because market volatility is also one of our state signals in Equation (5), we examine whether volatility alone explains the pricing-on/off pattern. For consistency with Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024), we expand the factor set to include the momentum (UMD) factor of Carhart (1997), the profitability (ROE) and investment (IA) factors of Hou, Xue, and Zhang (2015), and the betting-against-beta (BAB) factor of Frazzini and Pedersen (2014). For UMD, ROE, and IA, we follow standard constructions used in the factor literature. For BAB, we construct firm-level betas following Frazzini and Pedersen (2014) and compute SED without demeaning stock returns to preserve the common component, which is essential for timing BAB given its positive market exposure.¹⁵ Following DeMiguel et al. (2024), we sort sample months into terciles by realized market volatility and define low (high) volatility periods as months below the bottom (above the top) tercile.

Table IV compares pricing-on and pricing-off performance for eight factors across sample periods. Panel A reports results for the full sample. Over the all-time period, all factors earn significantly positive premia except SMB and HML, and BAB posts the highest Sharpe ratio (0.87). In their pricing-on periods, all factors display strong pricing power: pricing-on Sharpe ratios range from 0.60 (UMD) to 1.39 (BAB). In pricing-off periods, none of the factors exhibits pricing power, and average returns are insignificant across the board. Thus, our identification approach applies not only to the Fama-French factors but also to other widely studied factors. To quantify the improvement from identifying pricing-on/off periods, following Moreira and Muir (2017) we compute the appraisal ratio, defined as $\sqrt{SR_{\text{on}}^2 - SR_{\text{All}}^2}$. The gains are substantial. BAB records the highest ap-

¹⁵Specifically, UMD is based on stocks' cumulative return over the previous 12 months excluding the most recent month; ROE is income before extraordinary items divided by one-quarter-lagged book equity; and IA is the annual change in total assets divided by one-year-lagged total assets. For BAB, we compute SED as $SED_{BAB,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left[(r_{i,t})^2 - (r_{i,t} - \hat{r}_{BAB,i,t})^2 \right]$.

Table IV

Pricing-on/off of Representative Equity Factors

This table reports average monthly returns and Sharpe ratios for eight equity factors under our pricing-on/off classification. Panel A presents full-sample results; Panels B and C restrict the sample to low- and high-market-volatility months, respectively. We sort sample months into terciles by realized market volatility and define low (high) volatility periods as months below the bottom (above the top) tercile. For each factor and panel, we show “All-time,” “Pricing-on,” and “Pricing-off” means and Sharpe ratios; t -statistics are in parentheses. The Appraisal ratio, defined as $(SR_{on}^2 - SR_{all}^2)^{1/2}$, measures the improvement from conditioning on the pricing state. The sample spans January 1975 to December 2023.

		SMB	HML	RMW	CMA	UMD	ROE	IA	BAB
Panel A: Full sample									
All-time	Mean	0.23	0.25	0.35	0.25	0.52	0.52	0.31	0.87
		(1.80)	(1.63)	(3.35)	(2.63)	(2.78)	(4.60)	(3.27)	(4.89)
Pricing-on	SR	0.26	0.28	0.52	0.42	0.41	0.68	0.52	0.87
		(3.66)	(3.94)	(5.18)	(3.13)	(2.81)	(6.37)	(3.33)	(6.18)
Pricing-off	Mean	0.73	0.60	0.51	0.38	0.76	0.64	0.41	1.43
		(0.93)	(0.70)	(0.86)	(0.61)	(0.60)	(0.94)	(0.64)	(1.39)
	SR	-0.21	-0.29	0.04	0.02	0.21	0.07	0.12	0.17
		(-1.31)	(-1.10)	(0.19)	(0.14)	(0.75)	(0.19)	(1.10)	(0.73)
Appraisal ratio		-0.23	-0.30	0.04	0.03	0.17	0.06	0.25	0.19
		(-1.31)	(-1.10)	(0.19)	(0.14)	(0.75)	(0.19)	(1.10)	(0.73)
		0.89	0.64	0.69	0.44	0.43	0.65	0.38	1.08
Panel B: Low-market-volatility sub-sample									
All-time	Mean	0.01	0.35	0.26	0.12	0.88	0.62	0.25	1.23
		(0.06)	(1.78)	(2.36)	(1.15)	(4.82)	(5.24)	(2.38)	(6.50)
Pricing-on	SR	0.02	0.52	0.62	0.28	1.24	1.27	0.57	2.06
		(0.24)	(1.93)	(2.65)	(1.98)	(4.65)	(5.09)	(2.75)	(5.79)
Pricing-off	Mean	0.06	0.42	0.34	0.29	0.91	0.69	0.42	1.54
		(0.10)	(0.67)	(0.84)	(0.63)	(1.43)	(1.44)	(0.93)	(2.31)
	SR	-0.04	0.18	0.11	-0.16	0.85	0.25	-0.03	0.73
		(-0.12)	(0.50)	(0.82)	(-1.11)	(2.96)	(1.36)	(-0.20)	(5.00)
Appraisal ratio		-0.05	0.24	0.27	-0.38	1.07	0.51	-0.06	1.69
		(-0.12)	(0.50)	(0.82)	(-1.11)	(2.96)	(1.36)	(-0.20)	(5.00)
		0.10	0.42	0.56	0.57	0.71	0.68	0.73	1.06
Panel C: High-market-volatility sub-sample									
All-time	Mean	0.50	0.10	0.46	0.38	-0.19	0.28	0.40	0.20
		(2.09)	(0.30)	(2.14)	(2.07)	(-0.42)	(1.09)	(2.18)	(0.55)
Pricing-on	SR	0.48	0.08	0.49	0.50	-0.10	0.26	0.51	0.15
		(2.86)	(1.72)	(2.74)	(2.28)	(-0.14)	(1.66)	(1.98)	(2.17)
Pricing-off	Mean	1.05	0.63	0.55	0.56	-0.09	0.42	0.53	1.10
		(1.15)	(0.56)	(0.70)	(0.70)	(-0.05)	(0.45)	(0.62)	(0.75)
	SR	0.00	-0.45	0.29	0.03	-0.32	-0.06	0.15	-0.81
		(0.01)	(-1.04)	(0.76)	(0.16)	(-0.51)	(-0.09)	(0.74)	(-1.75)
Appraisal ratio		0.00	-0.37	0.25	0.05	-0.18	-0.05	0.25	-0.66
		(0.01)	(-1.04)	(0.76)	(0.16)	(-0.51)	(-0.09)	(0.74)	(-1.75)
		1.04	0.55	0.50	0.49		0.36	0.35	0.73

praisal ratio (1.08). Other notable appraisal ratios are SMB (0.89), RMW (0.69), ROE (0.65), HML (0.64), CMA (0.44), UMD (0.43), and IA (0.38). These improvements even

exceed those from the in-sample volatility-timing approach of Moreira and Muir (2017).¹⁶

Panel B reports the results for the low-market-volatility subsample and largely strengthens the full-sample message in Panel A. Unconditionally, most factors deliver higher average returns and Sharpe ratios in low-volatility months, especially UMD, ROE, IA, and BAB. In contrast, SMB earns little to no premium during low-market-volatility periods. The pricing-on/off split remains powerful: during pricing-on months, all factors post economically and statistically strong Sharpe ratios, while pricing-off months show no pricing power and often weak or negative average returns. The appraisal ratios are also sizable. BAB records the largest gain (1.06), with IA, UMD, and ROE close behind (around the 0.7 range), followed by RMW and CMA (roughly 0.55-0.60). HML improves modestly (about 0.42), and SMB barely benefits (about 0.10). Thus, even within tranquil markets, the pricing-state signal adds information beyond volatility timing.

Panel C reports the results for the high-market-volatility subsample and preserves the core pattern but amplifies the dispersion across factors. Unconditionally, performance deteriorates for most factors, UMD even turns negative and BAB hovers near zero, while SMB improves relative to the full sample. The pricing-state split becomes sharper: pricing-on months again exhibit strong and significant Sharpe ratios across most factors, whereas pricing-off months are near zero or negative, with UMD and BAB particularly weak when off. Appraisal ratios are also notable in this high-volatility state. SMB posts the highest gain (1.04), and BAB, HML, RMW, and CMA follow (roughly 0.49 – 0.73). ROE and IA show meaningful but smaller improvements (around 0.35).

As such, the on-off structure is remarkably stable: pricing-on months consistently deliver high Sharpe ratios, while pricing-off months do not, regardless of the volatility regime. Our pricing-state identification is not a proxy for market volatility, and it provides incremental timing information within both low- and high-market-volatility states.

Lastly, we examine the directional impact of the state predictors on the individual

¹⁶Moreira and Muir (2017) construct a volatility-managed factor by scaling the original factor by the inverse of its realized variance in the preceding month. They then run in-sample time-series regressions of each volatility-managed factor on the nonmanaged factor. They show that the appraisal ratios across factors are HML (0.20), RMW (0.41), CMA (0.08), UMD (0.88), ROE (0.80), IA (0.32), and BAB (0.66). However, there is no improvement for SMB.

factors' pricing power. Following the approach introduced in Section III.B, we regress each factor's next-month *SED* on its top five most important predictors selected via LASSO over the full sample. This allows us to quantify the economic magnitude of each signal's impact on expected factor pricing performance.

Table V
Determinants of Factors' Pricing State

This table examines the directional impact of the selected signals in determining individual factors' expected pricing performance. We adopt an in-sample approach by estimating an OLS regression where the dependent variable is next month's volatility-normalized cross-sectional mean squared forecast error difference ($SED_{t+1}/\sigma_{M,t}$). The independent variables are the top five most important predictors identified via LASSO for each respective factor. All regressors are cross-sectionally standardized to have a mean of zero and a standard deviation of one. The table reports estimated OLS coefficients, *t*-statistics, and the R^2 of the fitted regressions. We scale the raw *SED* by 10,000 prior to normalization, so the reported coefficients reflect changes in the volatility-adjusted pricing power (with the numerator expressed in units of percentage squared, %²). The sample spans January 1975 to December 2023.

Variable	Coeff.	<i>t</i> -stat	R^2	Variable	Coeff.	<i>t</i> -stat	R^2
SMB				HML			
Past 9-month SED	-6.55	-4.55	17.15	Past 1-month SED	2.55	4.86	8.33
Number of IPOs	-2.35	-3.62		Long term yield	2.48	3.39	
Past 12-month SED	8.27	5.90		Past 9-month SED	0.41	0.83	
Stock returns	4.21	6.61		Stock returns	1.97	4.03	
Past 1-month SED	3.38	4.85		Dividend price ratio	-2.05	-2.89	
RMW				CMA & IA			
Past 12-month SED	0.92	3.43	3.26	Past 1-month SED	1.14	4.53	8.74
Past 18-month SED	-0.78	-3.74		Past 9-month SED	-0.09	-0.35	
Past 6-month SED	-0.01	-0.07		Stock returns	0.95	4.06	
Stock returns	-0.19	-1.77		Net equity expansion	0.92	3.75	
Net equity expansion	0.03	0.31		Inflation	-0.69	-2.97	
UMD				ROE			
Dividend yield	-0.56	-0.71	1.40	Past 12-month SED	3.45	3.40	5.28
Stock returns	-1.11	-1.65		Past 18-month SED	-2.50	-3.92	
Past 3-month SED	-0.42	-0.52		Stock returns	-1.21	-3.52	
Default yield spread	-1.82	-2.33		Book-to-market ratio	-0.44	-1.30	
Past 9-month SED	-0.12	-0.15		Past 9-month SED	-0.74	-0.94	
BAB							
Past 1-month SED	16.23	2.66	2.45				
Number of IPOs	-7.22	-1.31					
corporate bond return	10.58	1.91					
Term spread	10.66	1.93					
Default return spread	-0.24	-0.04					

The first panel of Table V reports the regression results for the SMB factor. Among the five selected signals, “Number of IPOs” and “Past 9-month SED” significantly and negatively predict the SMB pricing state. The coefficients are economically large. A one-standard-deviation increase in the speculative sentiment proxy “Number of IPOs” is

associated with a 2.35 reduction in the expected volatility-normalized SED. By contrast, “Past 12-month SED” and “Stock returns” significantly and positively predict SMB’s pricing power. A one-standard-deviation increase in “Stock returns” increases the expected volatility-normalized SED by 4.21. Thus, aggregate market performance exerts a large, procyclical effect on SMB’s predictive accuracy.¹⁷

Across the remaining panels, the coefficients exhibit common structures and economically large effects. For HML, “Past 1-month SED” and “Long term yield” materially raise the expected volatility-normalized SED (coefficients of 2.55 and 2.48, respectively), whereas a higher “Dividend price ratio” lowers it by 2.05. State persistence varies meaningfully depending on the lag: RMW loads positively on “Past 12-month SED” (0.92) but negatively on “Past 18-month SED” (-0.78). The investment pricing states (CMA & IA) are clearly procyclical: a one-standard-deviation increase in “Stock returns” and “Net equity expansion” raises the expected volatility-normalized SED by 0.95 and 0.92, respectively, whereas higher “Inflation” lowers it by 0.69.

For UMD, the pricing state behaves countercyclically to aggregate returns and credit risk: “Stock returns” and “Default yield spread” enter negatively at -1.11 and -1.82, respectively, indicating that momentum pricing power weakens in rising or high-credit-risk markets. ROE exhibits a similar countercyclical loading on “Stock returns” (-1.21) and displays complex short-horizon dynamics, heavily weighting “Past 12-month SED” (3.45) while being penalized by “Past 18-month SED” (-2.50). For BAB, recent state persistence (“Past 1-month SED”, 16.23), “Term spread” (10.66), and “corporate bond return” (10.58) drastically increase expected volatility-adjusted pricing power, while speculative activity (“Number of IPOs”) reduces the expected volatility-normalized SED by 7.22. In terms of regression fit, the selected signals explain a substantial portion of the variation in future pricing power, with R^2 reaching 17.15% for SMB, 8.74% for CMA & IA, and 8.33% for HML.

¹⁷Small firms are harder to arbitrage (Shleifer and Vishny, 1997) and exhibit slower information diffusion (Hong, Lim, and Stein, 2000). Thus, small stocks, relative to large stocks, often display delayed price adjustment to aggregate information. Consistent with this view, Hou and Moskowitz (2005) find that small stocks exhibit greater price delay to aggregate shocks, as proxied by market returns.

V. Mean-Variance Efficient Portfolios

The preceding sections document a fundamental empirical regularity: factor pricing power is episodic, with risk premia concentrated in identifiable “pricing-on” states. This finding challenges the conventional assumption of constant factor exposures in portfolio construction. If the risk-return trade-off of a factor deteriorates significantly during “pricing-off” periods—as shown in our analysis of individual factors and models—a rational investor should dynamically adjust their allocation rather than holding a static portfolio. In this section, we quantify the economic magnitude of episodic factor pricing by examining the performance of mean-variance efficient portfolios that condition on these real-time states.

A. Multifactor Portfolios

We extend the analysis to a multifactor setting by forming a portfolio of the factor excess returns. We choose weights to obtain the mean-variance efficient (MVE) portfolio in factor space. This portfolio spans and thus prices the factors. Thus, to the extent that these factors price a broad set of assets and strategies, the MVE factor portfolio also prices that broader set. Specifically, we optimize factor weights by maximizing the mean-variance utility of an investor with risk aversion parameter γ :

$$\max_{\omega \geq 0} \omega^\top \hat{\mu} - \frac{\gamma}{2} \omega^\top \hat{\Sigma} \omega, \quad (11)$$

where $\hat{\mu}$ and $\hat{\Sigma}$ are the sample mean and covariance matrix of the factor-return vector, so $\omega^\top \hat{\mu}$ and $\omega^\top \hat{\Sigma} \omega$ are the sample mean and variance of the multifactor portfolio return. Following DeMiguel, Martín-utrera, and Uppal (2024), we impose the nonnegativity constraint $\omega \geq 0$ to mitigate estimation error and set $\gamma = 5$. We implement the MVE portfolio out of sample using an expanding window that begins with 60 months: the first estimation window spans January 1975 to December 1979 to form the portfolio held in January 1980, and we then extend the window by one month to form the February 1980 portfolio and continue until the sample ends. As such, the realized return of the multi-

factor portfolio (MP) at month t is: $r_{\text{MP},t} = \boldsymbol{\omega}_{t|t-1}^\top \mathbf{R}_t$, where $\mathbf{R}_t$ is a vector of factors' realized return at month t .

To consider our pricing-on/off identification, we construct a state-managed multifactor portfolio (SMP). Specifically, we continue to follow the above procedure of forming MP to choose portfolio weights. However, if we expect that a factor will be in its pricing-off state in the next month, we retain only a fraction of the holdings of that factor and liquidate the remaining fraction, with a retention ratio of η . Thus, the realized return of the SMP at month t is: $r_{\text{SMP},t} = \boldsymbol{\omega}_{t|t-1}^\top \text{diag}(\eta \mathbf{e} + (1 - \eta) \mathbf{I}_{t|t-1}) \mathbf{R}_t$, where $\mathbf{I}_{t|t-1}$ is the pricing-on/off indicator vector (each $I_i \in \{0, 1\}$, with $I_i = 1$ if pricing-on, and 0 if pricing-off), $\mathbf{e}$ is an all-ones vector, and η is a scalar parameter capturing the retention ratio. We apply the same retention ratio to all factors.

Table VI

Mean-Variance Efficient Multifactor Portfolios

This table reports the out-of-sample performance of mean-variance efficient multifactor portfolios that either ignore (MP) or exploit (SMP) factors' pricing states. We construct MVE portfolios by solving Problem (11). We consider different combinations of the nine factors in our dataset, including MKT, SMB, HML, RMW, CMA, UMD, ROE, IA, and BAB factors. For each specification, the table reports the out-of-sample monthly mean return (percent), the annualized Sharpe ratio, the p -value for the difference in Sharpe ratios between MP and SMP, and the appraisal ratio, defined as $(\text{SR}_{\text{SMP}}^2 - \text{SR}_{\text{MP}}^2)^{1/2}$, which measures the improvement from conditioning on the pricing state. Panel A covers the full out-of-sample period, and Panel B covers the post-2000 subperiod. The out-of-sample period spans from January 1980 to December 2023.

		FF3	FF3+UMD	FF5	FF5+UMD	Nine Factors
<i>Panel A: Full sample</i>						
MP	Mean	0.46 (2.87)	0.67 (4.50)	1.08 (5.25)	1.13 (6.12)	1.32 (6.30)
	SR	0.50	0.74	0.96	1.06	1.18
SMP	Mean	0.65 (5.53)	0.82 (6.94)	1.17 (7.23)	1.20 (7.64)	1.39 (9.43)
	SR	0.88	1.09	1.22	1.31	1.56
p -value ($\text{SR}_{\text{SMP}} - \text{SR}_{\text{MP}}$)		0.00	0.00	0.01	0.01	0.00
Appraisal ratio		0.72	0.80	0.75	0.78	1.01
<i>Panel B: After 2000</i>						
MP	Mean	0.47 (1.90)	0.49 (2.19)	1.06 (3.72)	1.00 (3.74)	0.92 (3.69)
	SR	0.43	0.48	0.85	0.84	0.80
SMP	Mean	0.57 (3.33)	0.68 (4.15)	1.04 (4.43)	1.05 (4.58)	0.96 (5.62)
	SR	0.68	0.84	1.02	1.08	1.10
p -value ($\text{SR}_{\text{SMP}} - \text{SR}_{\text{MP}}$)		0.02	0.01	0.10	0.04	0.01
Appraisal ratio		0.52	0.69	0.55	0.68	0.75

Table VI reports the performance of MVE multifactor portfolios. For each specification, the table reports the out-of-sample monthly mean return, the annualized Sharpe ratio, the p -value for the difference in Sharpe ratios between the pricing state-managed and unmanaged portfolios, and the appraisal ratio, which measures the improvement from conditioning on the pricing state. The significance of Sharpe ratio differences is assessed using the Jobson-Korkie test with the correction of Cederburg, O'Doherty, Wang, and Yan (2020).¹⁸ Portfolios are constructed using different combinations of the nine factors in our data set. We include the market factor in all combinations, although we manage only the non-market factors based on their cross-sectional pricing power.

Panel A shows that the pricing-state-managed portfolio delivers an out-of-sample Sharpe ratio that is significantly larger than the unmanaged portfolio across all specifications. For example, the unmanaged portfolio based on FF3 factors attains a Sharpe ratio of 0.50. By managing the pricing state of SMB and HML, we increase the Sharpe ratio by 76% to 0.88, and the difference is statistically significant at the 1% level. Adding factors improves out-of-sample performance for both unmanaged and managed portfolios. The unmanaged portfolio based on FF5 plus UMD reaches a Sharpe ratio of 1.06, and even at that level, timing the factors' pricing power raises the Sharpe ratio to 1.31, a 23.6% increase. The appraisal ratio remains economically meaningful at 0.78. Finally, we observe similarly large gains when we include all nine factors: the unmanaged portfolio records its highest Sharpe ratio of 1.18, while the managed portfolio further increases it to 1.56, a 32% improvement. The appraisal ratio also reaches its highest level at 1.01.

Panel B shows the out-of-sample performance after 2000. We can observe weaker unconditional performance but persistent gains from pricing-state management. Across all combinations, the unmanaged Sharpe ratios fall relative to Panel A (e.g., the nine-factor MP drops from 1.18 to 0.80). The managed portfolios continue to outperform, with Sharpe ratios rising to 0.68 (FF3), 0.84 (FF3+UMD), 1.02 (FF5), 1.08 (FF5+UMD), and

¹⁸Let $\hat{\mu}_i$ and $\hat{\sigma}_i$ denote the mean and standard deviation of excess returns for portfolio i over a period of length T , and let $\hat{\mu}_j$, $\hat{\sigma}_j$, and $\hat{\sigma}_{i,j}$ denote the corresponding quantities for portfolio j and the covariance between the two portfolios. To test the null hypothesis of equal Sharpe ratios for portfolios i and j , we compute the following Jobson and Korkie (1981) test statistic, which is asymptotically distributed as a standard normal: $\hat{z} = (\hat{\sigma}_j \hat{\mu}_i - \hat{\sigma}_i \hat{\mu}_j) / (\hat{\pi})^{1/2}$, where $\hat{\pi} = \frac{1}{T} \left(2\hat{\sigma}_i^2 \hat{\sigma}_j^2 - 2\hat{\sigma}_i \hat{\sigma}_j \hat{\sigma}_{i,j} + \frac{1}{2}\hat{\mu}_i^2 \hat{\sigma}_j^2 + \frac{1}{2}\hat{\mu}_j^2 \hat{\sigma}_i^2 - \frac{\hat{\mu}_i \hat{\mu}_j}{\hat{\sigma}_i \hat{\sigma}_j} \hat{\sigma}_{i,j}^2 \right)$.

1.10 (nine factors). Improvements are statistically significant across all specifications. Appraisal ratios remain economically meaningful, ranging from 0.52 to 0.75, though below Panel A (0.72 to 1.01), with the highest value again for the nine-factor specification. Overall, even in the post-2000 sample, the pricing-state signal mitigates the decline in unconditional performance and delivers sizable, often statistically significant improvements, comparable in relative terms to those in Panel A.

B. *Cost-aware Multifactor Portfolios*

Table VI shows that managing the factors' pricing states can substantially improve the out-of-sample performance of multifactor portfolios. However, actively timing the pricing state raises turnover and trading costs, which may erode these gains in practice. To examine the impact of transaction costs on our findings, we consider cost-aware MVE multifactor portfolios.

Following DeMiguel, Martín-utrera, and Uppal (2024), we construct multifactor portfolios by maximizing an investor's net-of-transaction-costs mean-variance utility with risk aversion γ ,

$$\max_{\boldsymbol{\omega} \geq 0} \boldsymbol{\omega}^\top \widehat{\boldsymbol{\mu}} - \widehat{\mathbf{TC}}(\boldsymbol{\omega}) - \frac{\gamma}{2} \boldsymbol{\omega}^\top \widehat{\boldsymbol{\Sigma}} \boldsymbol{\omega}, \quad (12)$$

where $\widehat{\mathbf{TC}}(\boldsymbol{\omega})$ denotes the sample transaction cost of the multifactor portfolio. We maintain the nonnegativity constraint $\boldsymbol{\omega} \geq 0$ to mitigate estimation error and set $\gamma = 5$. To compute the sample transaction cost, we first obtain the vector of stock trades needed to rebalance the portfolio at time $t+1$,

$$\Delta \mathbf{W}_{t+1}(\boldsymbol{\omega}) = \mathbf{W}_{t+1}(\boldsymbol{\omega}) - \mathbf{W}_t(\boldsymbol{\omega})^*, \quad (13)$$

where

$$\mathbf{W}_{t+1}(\boldsymbol{\omega}) = \sum_{k=1}^K x_{k,t+1} \boldsymbol{\omega}_k = \mathbf{X}_{t+1} \boldsymbol{\omega}, \quad (14)$$

$$\mathbf{W}_t(\boldsymbol{\omega})^* = \mathbf{W}_t(\boldsymbol{\omega}) \circ (\mathbf{e} + \mathbf{r}_{t+1}), \quad (15)$$

are, respectively, the multifactor portfolio at time $t+1$ and the portfolio before rebalancing at time $t+1$. Here, $x_{k,t+1} \in \mathbb{R}^{N_{t+1}}$ denotes the stock portfolio associated with the k^{th} factor, K is the number of factors, N_{t+1} is the number of stock at time $t+1$, ω_k is the portfolio weight on the k^{th} factor, $\mathbf{X}_{t+1} \in \mathbb{R}^{N_{t+1} \times K}$ stacks the K vectors $x_{k,t+1}$, and $\mathbf{a} \circ \mathbf{b}$ denotes the Hadamard (elementwise) product.

Given an estimation window with T historical observations of stock returns and factor portfolios, we estimate the average rebalancing cost of the conditional multifactor portfolio as

$$\widehat{\text{TC}}(\boldsymbol{\omega}) = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\boldsymbol{\Lambda}_t \Delta \mathbf{W}_{t+1}(\boldsymbol{\omega})\|_1, \quad (16)$$

where $\|\mathbf{a}\|_1$ denotes the 1-norm of the N_t -dimensional vector $\mathbf{a}$, and the transaction-cost matrix at time t , $\boldsymbol{\Lambda}_t$, is the diagonal matrix whose i^{th} diagonal element contains the transaction-cost parameter $\kappa_{i,t}$ of stock i at time t . As noted by DeMiguel, Martín-utrera, and Uppal (2024), the transaction-cost term in Equation (16) accounts for netting rebalancing trades across factors: we first net trades across the K factor portfolios and then apply transaction costs at the individual-stock level. Because netting reduces trading, multifactor portfolios benefit more when turnover is higher. We take the estimated transaction-cost parameter κ and individual-stock portfolio weights for the nine factors from DeMiguel, Martín-utrera, and Uppal (2024), which restricts our analysis to the period ending in December 2020.¹⁹

To incorporate pricing-state identification into the optimization, we replace $x_{k,t+1}$ with $x_{k,t+1} \cdot (\eta + (1 - \eta) \mathbf{I}_{k,t+1|t})$, where $\mathbf{I}_{k,t+1|t} \in (0, 1)$ is the pricing-on/off indicator ($\mathbf{I}_{k,t+1|t} = 1$ if factor k is in the pricing-on state at time $t+1$, and 0 if pricing-off). Thus, when factor k is expected to be pricing-off at $t+1$, we liquidate a $1 - \eta$ fraction of the current holdings associated with factor k , retaining only an η share. The resulting pricing-state-managed multifactor portfolio is optimized net of transaction costs.

¹⁹DeMiguel, Martín-utrera, and Uppal (2024) estimate the stock-level transaction-cost parameter $\kappa_{i,t}$ by using the two-day corrected method proposed by Abdi and Ranaldo (2017) to estimate the monthly bid-ask spread of the i^{th} stock as: $\hat{s}_{i,t} = \frac{1}{D} \sum_{d=1}^D \hat{s}_{i,d}$ and $\hat{s}_{i,d} = \sqrt{\max\{4(\text{cls}_{i,d} - \text{mid}_{i,d})(\text{cls}_{i,d} - \text{mid}_{i,d+1}), 0\}}$, where D is the number of days in month t , $\hat{s}_{i,d}$ is the two-day bid-ask spread estimate, $\text{cls}_{i,d}$ is the closing log-price on day d , and $\text{mid}_{i,d}$ is the mid-range log-price on day d ; that is, the mean of daily high and low log-prices. Because the effective trading cost is half the bid-ask spread, the transaction cost parameter for the i^{th} stock is $\kappa_{i,t} = \hat{s}_{i,t}/2$.

Table VII

Net-of-costs Performance of Multifactor Portfolios

This table evaluates cost-aware multifactor portfolios that either ignore (MP) or exploit (SMP) factors' pricing states, as well as an MVE combination of the two sleeves (SMP⁺). The portfolios are constructed using all nine factors in our dataset. We construct cost-aware MVE portfolios by solving Problem(12). To build SMP⁺, we first form cost-aware MP and SMP sleeves and then use the optimizer in Equation(11) to form the MVE combination, which assigns optimal weights to the two sleeves. This procedure effectively trades 17 factor portfolios simultaneously. For each portfolio, the table reports the out-of-sample net-costs mean, the annualized Sharpe ratio of net returns (accounting for trading diversification), and the p -value for the difference between the Sharpe ratios of managed and unmanaged portfolios. The table also reports the appraisal ratio, which measures the improvement from conditioning on the pricing state, and the average trading cost (TC), expressed as a fraction of the gross mean return. We report out-of-sample performance from January 1980 to December 2020.

	MP	SMP					
		Retention ratio, η					
		0.0	0.1	0.2	0.3	0.4	0.5
<i>Panel A: SMP with managed factors only</i>							
Mean	0.78 (4.31)	0.84 (5.01)	0.86 (5.12)	0.88 (5.13)	0.88 (5.13)	0.87 (5.01)	0.87 (4.98)
SR	0.94	1.02	1.05	1.06	1.07	1.06	1.06
p -value ($SR_{SMP} - SR_{MP}$)		0.21	0.11	0.06	0.03	0.02	0.00
Appraisal ratio		0.40	0.48	0.50	0.52	0.49	0.49
TC	0.24	0.40	0.38	0.36	0.34	0.32	0.30
<i>Panel B: SMP⁺ with 17 all factors</i>							
Mean	0.78 (4.31)	0.90 (4.98)	0.91 (5.05)	0.90 (5.02)	0.89 (4.96)	0.88 (4.83)	0.88 (4.78)
SR	0.94	1.09	1.10	1.09	1.09	1.07	1.06
p -value ($SR_{SMP^+} - SR_{MP}$)		0.04	0.02	0.01	0.01	0.02	0.01
Appraisal ratio		0.55	0.57	0.56	0.55	0.51	0.50
TC	0.24	0.33	0.32	0.31	0.30	0.30	0.29

Table VII reports the out-of-sample and net-of-costs performance of pricing-state managed (SMP) and unmanaged (MP) multifactor portfolios. We focus on nine-factor specifications, which benefit most from trading diversification, and evaluate SMP portfolios with retention ratios from 0 to 0.5. The table lists the net mean return, annualized Sharpe ratio, and the p -value for the Sharpe ratio difference between SMP and MP. The table also reports the appraisal ratio, which measures the improvement from conditioning on the pricing state, and the average trading cost (TC), expressed as a fraction of the gross mean return.

Panel A manages all nine factors except the market. As expected, MP's Sharpe ratio declines to 0.94 after transaction costs, which absorb, on average, 24% of gross profits.

SMP delivers higher net-of-costs performance. With $\eta = 0$, SMP attains a Sharpe ratio of 1.02, 8.5% above MP, along with an economically meaningful appraisal ratio of 0.40. The gains are partially offset by trading costs, which average 40% of the gross mean return, stemming from high turnover. Increasing the retention ratio lowers turnover and costs and raises performance. For example, with $\eta = 0.1$, trading costs fall to 38% of the gross mean return, and the Sharpe ratio rises to 1.05, an 11.7% improvement. With $\eta = 0.5$, trading costs drop to 30% and the Sharpe ratio rises to 1.06, and the difference versus MP is statistically significant at the 1% level.

Motivated by the trading-diversification effect in DeMiguel, Martín-utrera, and Uppal (2024), we also construct a multifactor portfolio, SMP^+ , that trades MP and SMP jointly. To mitigate estimation error from a large factor set, we first build cost-aware MP and SMP sleeves (i.e., independently managed sub-portfolios) as in Panel A. We then form the MVE portfolio SMP^+ of MP and SMP using the optimizer in Equation (11), which assigns optimal weights to the two sleeves. This approach effectively trades 17 factor portfolios jointly.²⁰ We compute the trading costs of SMP^+ by netting rebalancing trades across all 17 factor portfolios and then charging costs at the stock level, as in Equation (16).

Panel B shows that SMP^+ further improves performance. In particular, with $\eta = 0$, SMP^+ achieves a Sharpe ratio of 1.09 (16% above MP), with a p -value for the difference of 0.04 and an appraisal ratio of 0.55. The improvement over SMP reflects lower trading costs (33% versus 40%), indicating additional benefits from trade netting when unmanaged factors are included. Results are similar for other retention ratios. The highest Sharpe ratio of 1.10 occurs at $\eta = 0.1$ and significantly exceeds MP's 0.94. This specification also delivers the highest appraisal ratio of 0.57.

Figure 5 plots the cumulative out-of-sample net returns of the MP and SMP^+ with $\eta = 0$, with both portfolios standardized to have an annualized volatility of 10%. Evidently, the managed portfolio outperforms throughout the sample, particularly after 1985, and substantially mitigates the 1998 drawdown.

²⁰The factor portfolios include the market factor, the eight pricing-state-managed factors examined in Table IV, and their eight unmanaged raw counterparts.

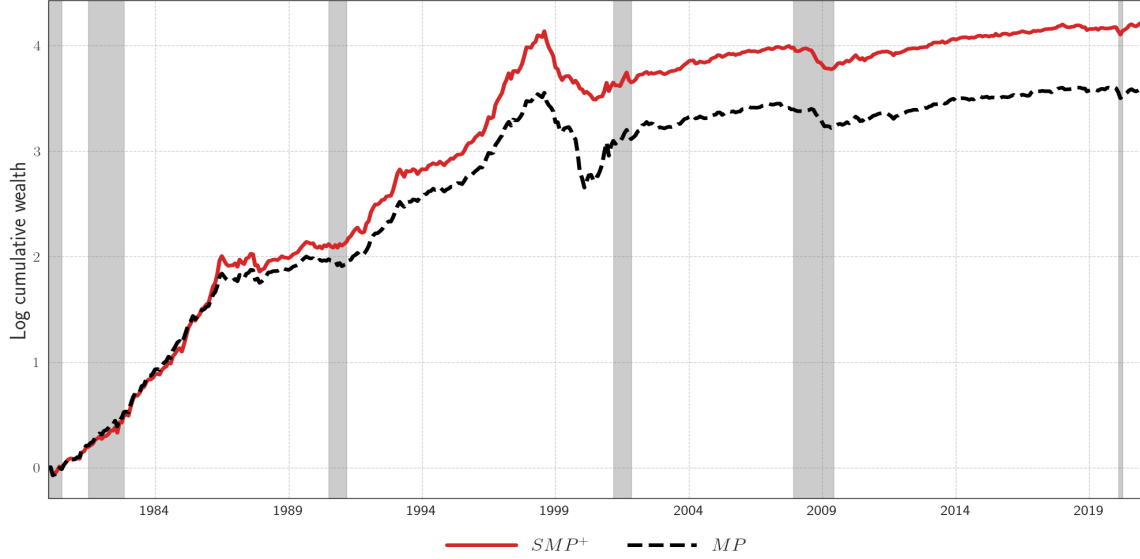


Figure 5. Net-costs cumulative wealth of multifactor portfolios. This figure depicts the out-of-sample cumulative value net of transaction costs of the mean-variance multifactor portfolios. To construct the augmented pricing-state-managed portfolio, SMP^+ , we first build cost-aware multifactor portfolios that either ignore (MP) or exploit (SMP) factors’ pricing state by solving Problem (12), and then form the MVE portfolio SMP^+ based on MP and SMP using the optimizer in Equation (11), which assigns optimal weights to the two sleeves. This procedure effectively trades 17 factor portfolios simultaneously. The initial investment is \$1, and both portfolios are standardized to have the same annualized volatility of 10%. The out-of-sample period spans from January 1980 to December 2020. The shaded areas denote NBER recessions.

C. Economic Implications

Panel B of Figure 3 shows that the risk-return trade-off is substantially stronger when factors are pricing-on and weakens sharply in the pricing-off state.²¹ A mean-variance investor should therefore scale down factor exposures during pricing-off periods. Consistent with this mechanism, our pricing-state-managed multifactor portfolio, which reduces risk-factor exposures when they are pricing-off, outperforms its unmanaged counterpart.

To assess the economic implication of our findings, we follow DeMiguel, Martín-utrera, and Uppal (2024) and estimate a conditional SDF whose factor prices of risk vary with each factor’s pricing state. We assume that, conditional on the factors’ pricing states, an SDF prices traded assets, that is, $0 = E_t(M_{t+1}^I r_{t+1}^e)$, where M_{t+1}^I is the conditional SDF at time $t + 1$ and r_{t+1}^e is the vector of excess asset returns at time $t + 1$. We further

²¹A similar pattern holds across other factors. Specifically, the risk-return trade-off for UMD, ROE, IA, and BAB is 38.46 ($t = 4.93$), 30.29 ($t = 3.69$), 53.04 ($t = 6.70$), and 57.33 ($t = 6.51$), respectively, during their pricing-on state, but only 23.67 ($t = 2.85$), 8.59 ($t = 1.08$), 5.56 ($t = 0.65$), and 23.91 ($t = 2.92$), respectively, during pricing-off states.

assume that the price of risk for factor k is affine in its pricing state, so that

$$M_{t+1}^I = 1 - \sum_{k=1}^K (\alpha_k + \beta_k \tilde{I}_{k,t+1|t}) \tilde{r}_{k,t+1}, \quad (17)$$

where $\tilde{I}_{t+1|t}$ is the centered time- $t+1$ pricing-state indicator available at time t , $\tilde{I}_{k,t+1|t} = I_{k,t+1|t} - E(I_k)$, I_k is the pricing-on/off indicator ($I_k \in \{0, 1\}$ equals 1 in pricing-on state and 0 in pricing-off state), and $\tilde{r}_{k,t+1}$ is the conditionally demeaned k^{th} factor return at time $t+1$, $\tilde{r}_{k,t+1} = r_{k,t+1} - E(r_{k,t+1})$. The SDF prices every traded factor return, and thus

$$E_t(r_{i,t+1}) = \sum_{k=1}^K (\alpha_k + \beta_k \tilde{I}_{k,t+1|t}) \text{cov}_t(r_{i,t+1}, r_{k,t+1}), \quad \text{for } i = 1, 2, \dots, K, \quad (18)$$

where $\text{cov}_t(r_{i,t+1}, r_{k,t+1})$ is the conditional covariance between the i^{th} and k^{th} factor returns.

To estimate the coefficients α_k and β_k , one could run a pooled conditional regression across the K equations in (18), but doing so would require estimating $K(K+1)/2$ realized factor variances and covariances each month. Instead, following DeMiguel, Martín-utrera, and Uppal (2024), we adopt a parsimonious and robust approach that ignores cross-factor correlations:

$$E_t(r_{k,t+1}) = (\alpha_k + \beta_k \tilde{I}_{k,t+1|t}) \text{var}_t(r_{k,t+1}), \quad \text{for } k = 1, 2, \dots, K, \quad (19)$$

which requires estimating only K realized factor-return variances each month. Moreover, Equation (19) implies that, when factor returns are uncorrelated, the conditional risk-return trade-off for the k^{th} factor is equal to the price of risk for the k^{th} factor:

$$\frac{E_t(r_{k,t+1})}{\text{var}_t(r_{k,t+1})} = \alpha_k + \beta_k \tilde{I}_{k,t+1|t}, \quad \text{for } k = 1, 2, \dots, K. \quad (20)$$

To examine whether the price of risk or risk premium (RP) for factor k varies with

its pricing state, we estimate the time-series regression

$$RP_{k,t+1} = \frac{r_{k,t+1}}{\sigma_{k,t}^2} = \alpha_k + \beta_k \tilde{I}_{k,t+1|t} + \varepsilon_{k,t+1}, \quad (21)$$

where $\sigma_{k,t}^2$ denotes the monthly realized variance of the k^{th} factor, computed as the sample variance of its daily returns in month t , and $\varepsilon_{k,t+1}$ is the residual at time $t + 1$. The unconditional price of risk for the k^{th} factor is $E(r_{k,t+1}/\sigma_{k,t}^2) = E(\alpha_k + \beta_k \tilde{I}_{k,t+1|t} + \varepsilon_{k,t+1}) = \alpha_k$ because $E(\tilde{I}_{k,t+1|t}) = 0$ and $E(\varepsilon_{k,t+1}) = 0$. Accordingly, we test whether the unconditional price of risk is positive using the t -statistic for the estimated coefficient α_k , and we test for state dependence using the t -statistic for the estimated coefficient β_k .

Table VIII
State-Dependent Factor Risk Prices

This table examines the time-variation of factor risk prices. Panel A reports the coefficients α_k and β_k for the time-series regression defined in Equation (21) for the eight factors in our data set. Panel B augments the regression with a demeaned inverse realized market volatility control. The numbers in parentheses are Newey-West t -statistics. The sample spans January 1975 to December 2023.

	SMB	HML	RMW	CMA	UMD	ROE	IA	BAB
<i>Panel A: $RP_{k,t+1} = \alpha_k + \beta_{k,\tilde{I}} \cdot \tilde{I}_{k,t+1 t} + \varepsilon_{k,t+1}$</i>								
α_k	6.79 (1.65)	10.61 (1.60)	30.02 (4.48)	15.77 (2.54)	31.92 (5.33)	22.54 (3.80)	43.43 (6.63)	42.44 (6.59)
$\beta_{k,\tilde{I}}$	22.05 (3.09)	25.80 (2.08)	48.41 (4.22)	26.89 (2.12)	14.79 (1.39)	21.70 (1.84)	47.48 (3.91)	33.42 (3.66)
<i>Panel B: $RP_{k,t+1} = \alpha_k + \beta_{k,\tilde{I}} \cdot \tilde{I}_{k,t+1 t} + \beta_{k,\tilde{\sigma}^{-1}} \cdot \tilde{\sigma}_t^{-1} + \varepsilon_{k,t+1}$</i>								
α_k	6.79 (1.66)	10.61 (1.60)	30.02 (4.59)	15.77 (2.54)	31.92 (5.67)	22.54 (3.83)	43.43 (6.83)	42.44 (7.33)
$\beta_{k,\tilde{I}}$	22.05 (3.09)	22.48 (1.81)	50.28 (4.34)	26.78 (2.11)	15.83 (1.54)	21.50 (1.83)	44.12 (3.75)	25.48 (2.97)
$\beta_{k,\tilde{\sigma}^{-1}}$	0.19 (0.45)	0.85 (1.90)	1.35 (2.91)	0.43 (1.03)	1.71 (3.59)	0.79 (2.04)	1.24 (2.34)	2.34 (5.82)

Panel A in Table VIII estimates the time-series regression of each factor’s price of risk on a centered, factor-specific “pricing on/off” indicator. With this centering, β_k is exactly the on-off spread: the conditional mean in the on state minus that in the off state. Hence $\beta_k > 0$ means the price of risk is higher when the factor is pricing-on and lower when it is pricing-off.

The estimates show that the unconditional price of risk is positive for all eight factors.

α is 6.79 for SMB, 10.61 for HML, 30.02 for RMW, 15.77 for CMA, 31.92 for UMD, 22.54 for ROE, 43.43 for IA, and 42.44 for BAB. Statistically, α is significant for RMW, CMA, UMD, ROE, IA, and BAB (t -statistics between about 2.5 and 6.6), and only marginally significant for SMB and HML. Economically, IA and BAB have the largest unconditional prices of risk, followed by UMD and RMW.

Importantly, state dependence is strong. $\beta_{\hat{I}}$ is positive for every factor—22.05 (SMB), 25.80 (HML), 48.41 (RMW), 26.89 (CMA), 14.79 (UMD), 21.70 (ROE), 47.48 (IA), and 33.42 (BAB)—meaning the price of risk is higher in pricing-on months and attenuates in pricing-off months. Most $\beta_{\hat{I}}$ estimates are statistically significant (except for UMD).

DeMiguel, Martín-utrera, and Uppal (2024) find that the conditional price of risk for these factors decreases with realized market volatility. Thus, Panel B further augments the regression with a demeaned inverse realized market volatility as the control, $\tilde{\sigma}_t^{-1} = 1/\sigma_t - E(1/\sigma_t)$, where σ_t is the market volatility at time t , so $RP_{k,t+1} = \alpha_k + \beta_{k,\tilde{I}} \cdot \tilde{I}_{k,t+1|t} + \beta_{k,\tilde{\sigma}^{-1}} \cdot \tilde{\sigma}_t^{-1} + \varepsilon_{k,t+1}$. Because both regressors are mean zero, α_k remains the unconditional price of risk and is unchanged relative to Panel A. The $\beta_{\tilde{\sigma}^{-1}}$ estimates are positive for all factors. This pattern confirms the literature’s finding that the conditional price of risk rises when volatility is low and falls when volatility is high.²² After controlling for market volatility, the on/off coefficient $\beta_{\hat{I}}$ remains positive for every factor, indicating a higher risk-return trade-off in pricing-on months. Thus, Panel B shows two distinct effects on risk prices: a market volatility effect and a factor-specific pricing-on/off effect.

Taken together, the unconditional price of risk is broadly positive across standard factors, and more importantly, the price of risk is state dependent. When the indicator signals that the factor is “on,” the risk-return trade-off is materially stronger, and when it is “off,” the trade-off weakens, sometimes substantially. This on-off structure provides a clear and economically intuitive way to condition the SDF on factor-specific pricing states.

²²Moreira and Muir (2017) show investors can improve performance by reducing exposure to risk factors when their volatility is high. DeMiguel, Martín-utrera, and Uppal (2024) show that factor risk prices generally decrease with market volatility. Volatility timing also works for UMD and BAB factors, as shown by Barroso and Santa-Clara (2015), Cederburg and O’Doherty (2016), and Barroso, Detzel, and Maio (2025).

VI. Conclusion

This paper challenges the standard assumption that factor pricing power is stable over time. Instead, we show that factor pricing is inherently episodic. Using a real-time, out-of-sample framework to identify pricing-on and pricing-off states, we document that both factor premia and cross-sectional return predictability are concentrated almost entirely in active regimes while becoming weak or indistinguishable from zero during inactive periods.

The economic consequences of this state dependence are substantial. Sharpe ratios differ sharply across pricing states, indicating that the risk-return trade-off faced by investors varies meaningfully over time. A stochastic discount factor with state-dependent prices of risk provides a markedly improved characterization of the investment opportunity set relative to a constant-parameter alternative. Moreover, conditioning portfolio construction on pricing states yields economically large and statistically significant out-of-sample gains for mean-variance efficient portfolios, even after accounting for transaction costs.

Taken together, our results point to a simple but powerful on-off structure in asset pricing. Rather than reflecting a failure of factor models or a proliferation of spurious anomalies, weak factor performance often arises because pricing power itself is temporarily inactive. This perspective has important implications for the evaluation of factor premia, the interpretation of risk-return trade-offs, and the design of empirical asset-pricing tests. Because factor models are widely used across many areas of finance, extending the analysis of episodic pricing to other markets and asset classes represents a promising direction for future research.

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Internet Appendix of “Episodic Factor Pricing”

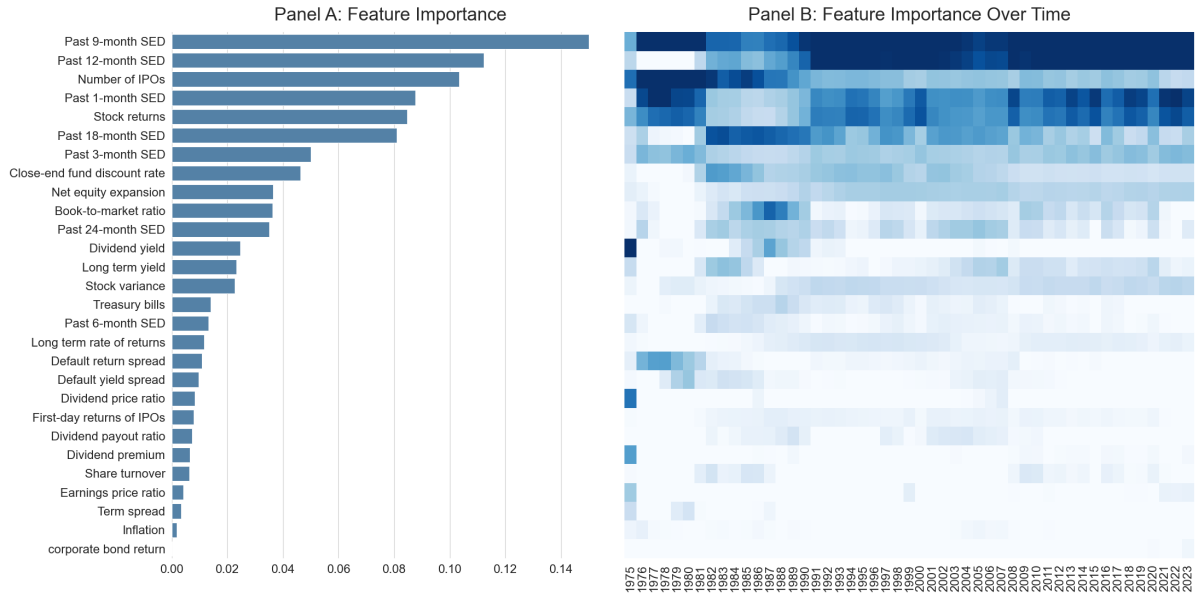


Figure IA.1. Feature importance for the FF5 model. This figure illustrates the relative importance of state predictors in forecasting the FF5 model’s *SED*. Panel A reports the relative contribution of each predictor, calculated as the time-series average of its absolute LASSO coefficient divided by the sum of all averaged absolute coefficients across all predictors. Panel B displays a heatmap of these relative contributions over time, calculated as the annual average of the normalized absolute coefficients. Darker colors indicate a higher relative contribution in a given year. The sample spans January 1975 to December 2023.

IA.A. Evidence From A Large Set of Factors

A. *On-Off Factor Performance*

This section extends the analysis to 211 factors drawn from the comprehensive online dataset of Chen and Zimmermann (2022). For each factor, we identify its pricing state based on its associated firm characteristic using the procedure in Section II.

Panel A in Figure IA.2 shows substantial time variation in the number of factors that are in a pricing-on state. On average, there are 125 pricing-on factors at any point in time, with a minimum of 84 and a maximum of 162. Thus, pricing-on states are common rather than rare, and during normal expansions, the series typically fluctuates in the 120 to 145 range.

The series is clearly procyclical. It falls markedly during NBER recessions and recovers in expansions. Notable troughs coincide with the early-1980s double-dip recession, the 1990/91 downturn, and especially the 2008 financial crisis, which delivers the deepest and most persistent decline to the sample low of 84. The 2001 recession produces a milder drawdown, while the 2020 COVID recession features a sharp but brief dip followed by a quick rebound. In several episodes, the deterioration begins shortly before the official recession start and bottoms near the end or shortly after, suggesting weak leading but stronger coincident/lagging behavior relative to the business cycle. Peaks occur in the late 1990s to early 2000s, the mid-2000s before the crisis, and 2013 to 2015, reaching about 155 to 162 factors. Thus, the strength of cross-sectional pricing varies broadly with macroeconomic conditions.

Our result indicates that recessions feature fewer pricing-on factors and a weaker overall risk-return trade-off. This finding is consistent with Moreira and Muir (2017) and DeMiguel, Martín-utrera, and Uppal (2024), who find that the risk-return trade-off weakens when market volatility is high. Therefore, our findings challenge structural models of time-varying expected returns, which typically predict a stronger market risk-return trade-off during recessions.

Panel B in Figure IA.2 reports the composition of pricing-on breadth across six factor

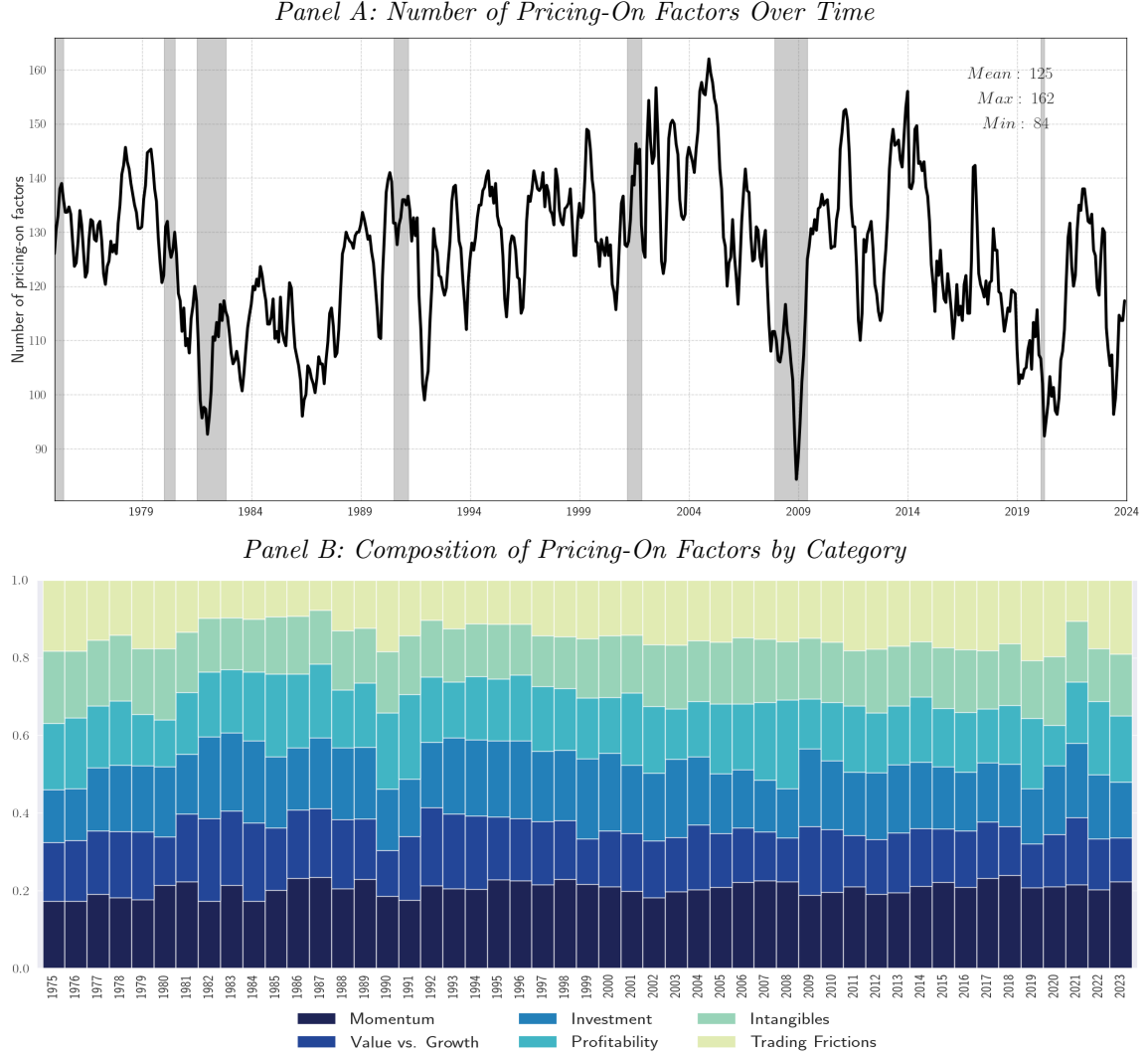


Figure IA.2. Time variation of breadth and composition of pricing-on factors. This figure summarizes the breadth and composition of pricing-on factors among 211 factors. Panel A plots the time series of the number of factors in the pricing-on state, with shaded areas denoting NBER recessions. Panel B shows the composition of pricing-on factors by category after normalization. For category c at time t , let $P_{c,t}$ be the number of pricing-on factors and $F_{c,t}$ the total number of factors in that category. Compute the within-category ratio $r_{c,t} = P_{c,t}/F_{c,t}$. Rescale across categories so the ratios sum to one, $s_{c,t} = r_{c,t}/\sum_c r_{c,t}$. The stacked bars plot the annual averages. This procedure controls for differences in category size and for changes in the overall level of pricing-on activity. The sample spans January 1975 to December 2023.

groups after within-group scaling and cross-group normalization, so each bar sums to one. Specifically, for category c at time t , let $P_{c,t}$ be the number of pricing-on factors and $F_{c,t}$ the total number of factors in that category. Compute the within-category ratio $r_{c,t} = P_{c,t}/F_{c,t}$. Rescale across categories so the ratios sum to one, $s_{c,t} = r_{c,t}/\sum_c r_{c,t}$. The stacked bars plot the annual averages. This procedure controls for differences in category size and for changes in the overall level of pricing-on activity. Evidently, momentum is the single biggest contributor in most periods. Across episodes, the mix rotates with

market regimes. Around the dot-com cycle (1999-2002), Momentum’s share narrows, while Profitability and, to a lesser extent, Intangibles take a larger slice. During and after the 2008 crisis, Investment and Profitability gain relative weight, while Value vs. Growth contracts. However, these shifts are modest compared with the large fluctuations in the overall count shown in Panel A. Thus, although there are significant changes in breadth, Panel B shows a fairly steady composition with episodic but not dramatic reweighting across categories.

Next, we examine whether the time-varying predictability of firm characteristics leads to differences in average returns for their corresponding factors. By analyzing the average returns of factors associated with different firm characteristics, we can assess the economic significance of their return predictability. If firm characteristics exhibit time-varying predictability, we would expect to observe varying average returns for the factors related to those characteristics over different periods.

Figure [IA.3](#) provides a comparison of average returns for 211 factors in their pricing-on vs. pricing-off state. Panel A presents the t -statistics of average factor returns for the top 50 factors with the longest pricing-on periods in our sample. Notably, for almost all factors, the average returns over pricing-on periods are both statistically and economically significant. The pricing-on t -statistics are above the threshold of 3 proposed by Harvey, Liu, and Zhu (2016) for most of the factors. In fact, for a large fraction of factors, the pricing-on t -statistics even exceed 4, indicating a high level of statistical significance. These findings support the economic value of timing investment strategies based on the factors’ pricing state. On the other hand, the pricing-off average returns are insignificantly different from zero for almost all factors. This reinforces the observation that the factor pricing power diminishes during pricing-off periods, making them less profitable for investors.

The histogram in Panel B contrasts the t -statistics of average returns for all 211 factors in pricing-on versus pricing-off states. The two distributions are clearly separated: the pricing-on t -statistics are shifted far to the right, with a mean around 3.62 (red dashed line), and exhibit a pronounced right tail, indicating that many factors exceed

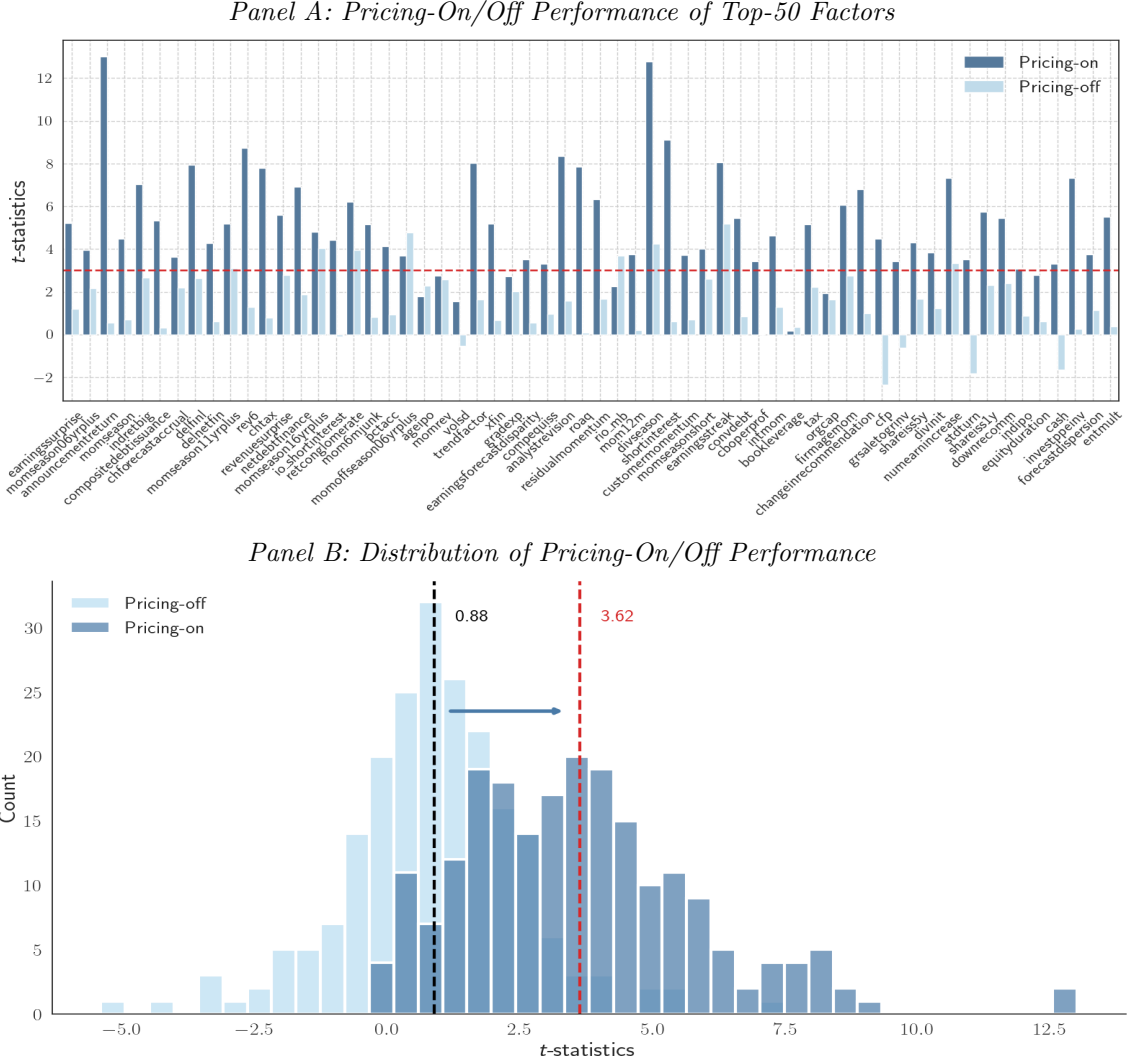


Figure IA.3. Pricing-on/off performance of 211 factors. This figure evaluates factor strength conditional on the pricing state. For each firm characteristic, we construct its factor return and partition the monthly sample into pricing-on and pricing-off months using the classification in Section II. For every characteristic and state, we compute the t -statistic of the mean factor return over the corresponding subset of months. Panel A reports the pair of state-specific t -statistics for the top 50 factors with the longest pricing-on periods. Panel B shows histograms of the state-conditional t -statistics for all 211 factors, with vertical lines indicating the cross-factor averages in each state. The sample spans January 1975 to December 2023.

the threshold of 3 and a sizable subset surpasses 4, with a few extreme values above 8. By contrast, the pricing-off distribution is centered near 0.88 (black dashed line), with mass concentrated between roughly -1 and 2. This figure highlights the large shift from the off to the on state, indicating that the same factors deliver statistically and economically meaningful average returns when they are “on,” but not when they are “off.” This pattern validates the pricing-state classification and underscores the investment value of timing exposures to factors only during their pricing-on periods.

B. On-Off Performance By Factor Category

In this section, we further examine the on-off performance for each factor category and ask which category benefits more from pricing-state timing.

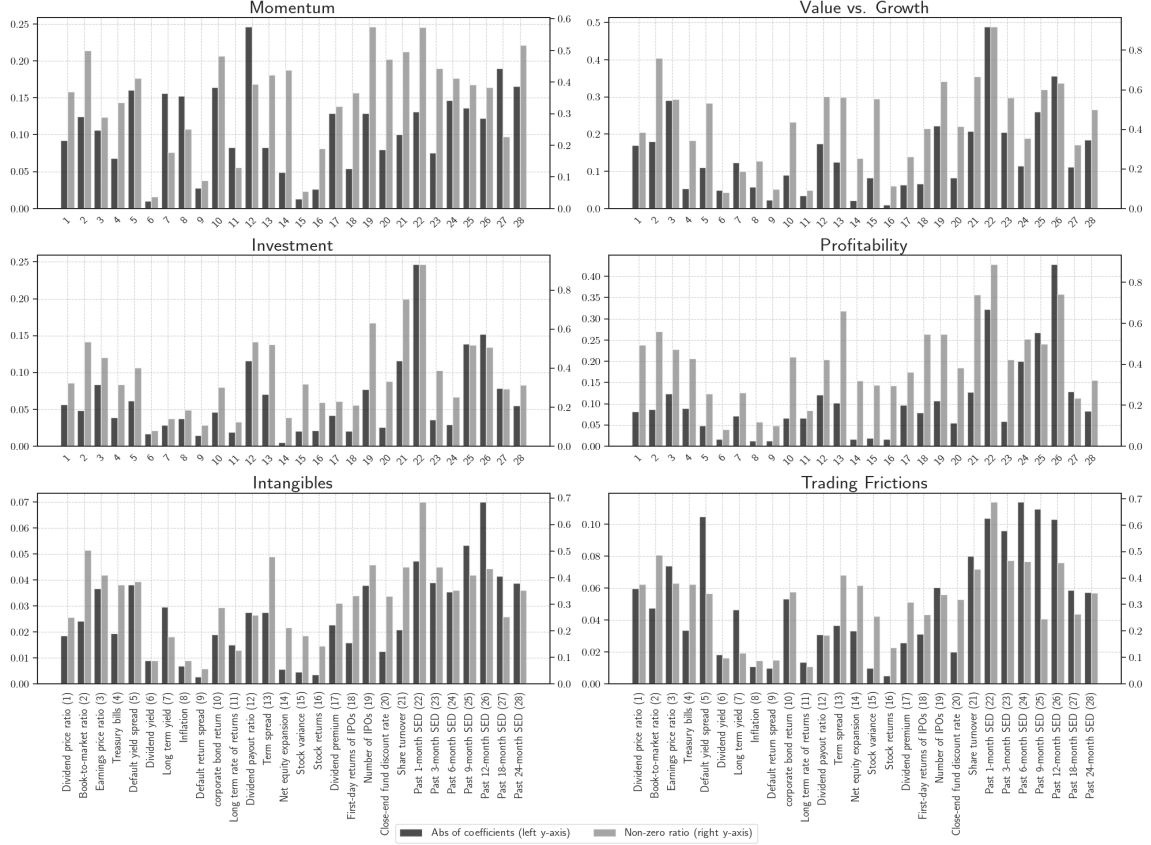


Figure IA.4. Feature importance in predicting factor pricing state. This figure summarizes the LASSO regressions, in Equation (5), for predicting each factor’s SED using market-wide and factor-specific variables. We compute two statistics for measuring its importance per factor: the time-series average of the absolute coefficient and the selection frequency (the fraction of months in which the LASSO keeps the coefficient non-zero). We average these two statistics across all factors within a category. Bars in each panel therefore report, on the left axis, average absolute coefficients and, on the right axis, the probability that a variable is selected. The sample spans January 1975 to December 2023.

We first examine feature importance for identifying the pricing-on/off state for each factor category. Figure IA.4 summarizes the LASSO regressions in Equation (5) for predicting each factor’s SED using market-wide and factor-specific variables. Each month, for every factor, we run the predictions as outlined in Section II. For a given predictor, we then compute two statistics for measuring its importance per factor: the time-series average of the absolute coefficient and the selection frequency (the fraction of months the LASSO keeps the coefficient non-zero). Finally, we average these two statistics across all factors within a category. Bars in each panel therefore report, on the left axis, average

absolute coefficients and, on the right axis, the probability that a variable is selected.

Several clear patterns emerge. Lagged SEDs are the dominant set of predictors across categories: their selection frequencies are the highest and their average magnitudes are large. Market-wide investor sentiment measures, such as dividend premium, IPO first-day returns, number of IPOs, closed-end fund discount, and share turnover, are frequently selected and economically important in Momentum, Value, and Trading Frictions, and to a lesser extent in Profitability and Intangibles. Market-wide variables, such as dividend-price, book-to-market, earnings-price, treasury bills, and default spreads, also matter in all categories. Investment leans heavily on past 1-month SED with comparatively limited use of macro state variables. Furthermore, although the mix of useful predictors is heterogeneous across categories, both market-wide and factor-specific variables are informative about factor pricing states.

Next, we compare performance across pricing-on and pricing-off states within each factor category. We scale all 211 factors to have an ex-ante annualized volatility of 10%. For each category, we form two portfolios based on the composite factors' pricing state, a pricing-on portfolio and a pricing-off portfolio. For category c at time t , the pricing on portfolio includes only composite factors predicted to be in the pricing-on state at time t . We use simple equal weighting. The return of the pricing-on portfolio at time t is $r_{\text{pricing-on},t}^c = \sum_i r_{i,t}^c \cdot I_{i,t} / N_t^c$, where $I_{i,t}$ equals 1 when factor i is in the pricing-on state and 0 otherwise, and N_t^c is the number of pricing-on factors in category c at time t .

Table [IA.I](#) shows that conditioning on the predicted pricing state substantially improves performance for every category. Mean returns and Sharpe ratios are uniformly higher in the pricing-on state, and the differences are statistically strong. Value exhibits the most dramatic contrast, with a mean of 0.75% and a Sharpe ratio of 1.83 when pricing-on, versus 0.09% and 0.20 when pricing-off, and very low t -statistics in the off state. Investment and Trading Frictions also post large gains, with Sharpe ratios of 1.68 and 1.72 when pricing-on and clear reductions when off. Momentum delivers a high mean of 0.85% and a Sharpe ratio of 1.46 when on, while the off state remains relatively resilient with a Sharpe ratio of 0.84. Intangibles records the highest pricing-on Sharpe ratio at 2.08

Table IA.I

Pricing-on/off Performance in Different Factor Categories

This table reports the performance for the pricing-on and pricing-off portfolios within each factor category. We first scale all 211 factors to an ex-ante annualized volatility of 10%. For each category and month, we form a pricing-on portfolio that equally weights the composite factors predicted to be in the pricing-on state and a pricing-off portfolio that equally weights those predicted to be in the pricing-off state. The table presents the mean monthly return with t -statistics in parentheses, the Sharpe ratio, and the p -value for the difference in Sharpe ratios. The sample spans January 1975 to December 2023.

	Momentum		Value		Investment		Profitability		Intangibles		Trading Frictions	
	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Mean	0.85	0.54	0.75	0.09	0.74	0.37	0.50	0.31	0.52	0.22	0.56	0.22
	(9.61)	(5.70)	(10.82)	(1.23)	(10.39)	(4.81)	(6.22)	(3.40)	(10.74)	(5.57)	(11.62)	(4.89)
SR	1.46	0.84	1.83	0.20	1.68	0.78	0.95	0.57	2.08	0.96	1.72	0.70
p -value ($SR_{on} - SR_{off}$)	0.00		0.00		0.00		0.01		0.00		0.00	

and maintains the strongest pricing-off Sharpe ratio at 0.96, which suggests more stable performance across states. Profitability shows a relatively smaller gain, with a pricing-on Sharpe ratio of 0.95 and a pricing-off value of 0.57. The p -values for the difference in Sharpe ratios reject equality for all categories. Comparing categories, the magnitude of state dependence varies meaningfully. Value, Trading Frictions, and Intangibles display the largest on-minus-off improvement. Investment shows substantial gains as well, driven by periods when its state signals turn on. Momentum and Profitability benefit notably from conditioning, although their off state does not deteriorate as much as others.

To examine the evolving cumulative returns of the two strategies over time, we calculate the cumulative profits for each strategy based on a monthly-rebalanced approach. Starting with an initial investment of $W_1 = 1$ at the beginning of January 1975, we track the trajectory of $W_t = W_{t-1} \times (1 + R_t)$ for each strategy. Figure IA.5 plots log cumulative wealth for pricing-on and pricing-off portfolios in each category starting from one dollar. In every panel, the pricing-on line compounds faster and stays above the pricing-off line.

Momentum shows steady and almost uninterrupted compounding in the on state through the sample, while the off state grows more slowly and flattens after the mid 2000s. Value displays the starkest split. The on state climbs almost linearly with a sharp surge around the late 1990s and another rise near the end of the sample, whereas the off state is largely flat and even drifts down in the 2010s. Investment also shows a wide gap.

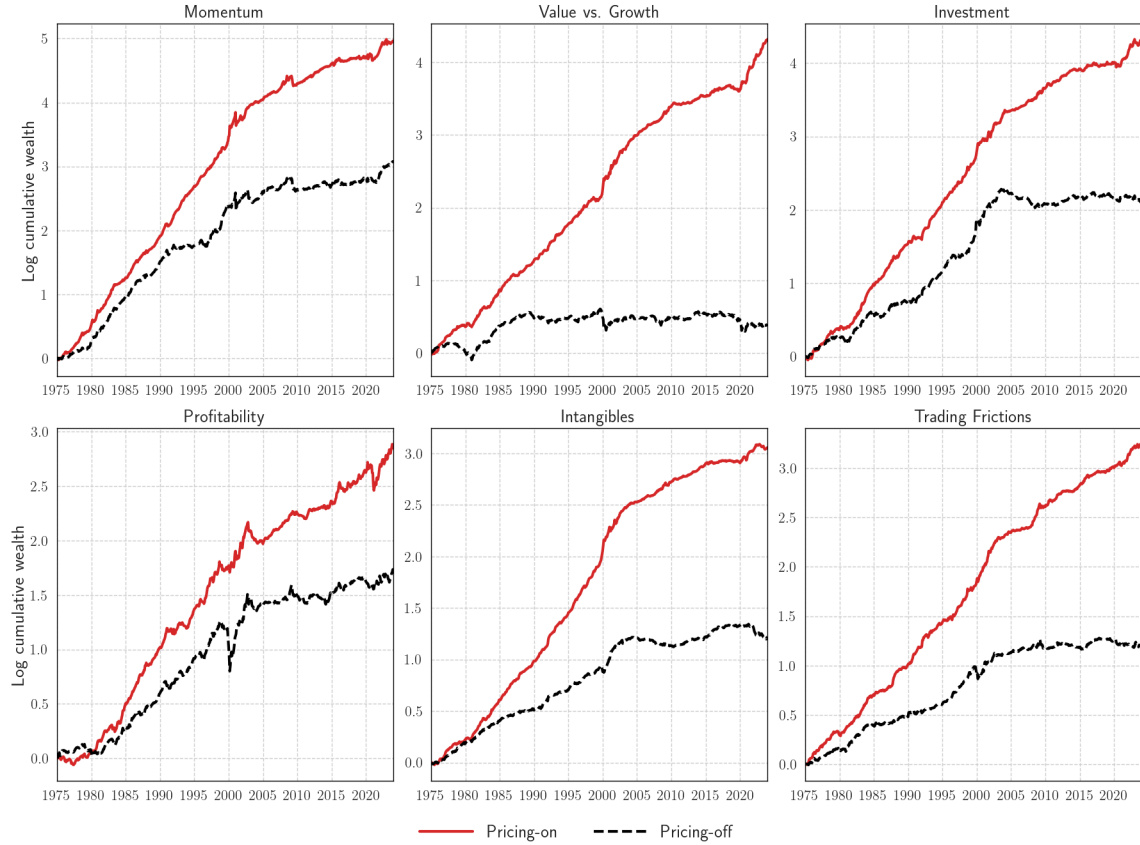


Figure IA.5. Cumulative wealth of category portfolios. This figure depicts the cumulative value of two factor category portfolios: pricing-on and pricing-off portfolios. We first scale all 211 factors to an ex-ante annualized volatility of 10%. For each category and month, we form a pricing-on portfolio that equally weights the composite factors predicted to be in the pricing-on state and a pricing-off portfolio that equally weights those predicted to be in the pricing-off state. We calculate the cumulative profits for each strategy based on a monthly rebalanced approach. Starting with an initial investment of $W_1 = 1$ at the beginning of January 1975, we track the trajectory of $W_t = W_{t-1} \times (1 + R_t)$ for each strategy. The sample spans January 1975 to December 2023.

The on state keeps rising after 2000, while the off state peaks early in the 2000s and then stagnates. Profitability exhibits the smallest separation. Both states trend upward with similar shapes and moderate drawdowns, but the on state remains consistently higher. Importantly, the pricing-on portfolio largely mitigates the large drawdown observed in the pricing-off portfolio around 2000. Intangibles rises rapidly in the on state from the mid 1980s to the early 2000s and then grows more gradually, while the off state accumulates only modestly. Trading Frictions shows strong step ups in the on state in the mid 1990s and again after the mid 2000s, while the off state levels off after 2000.

In sum, these results show that conditioning on the pricing state materially improves factor performance across all categories, although the magnitude of the gains differs by category. Value, Trading Frictions, and Intangibles benefit the most. The benefits are

especially pronounced after 2000 for all factor categories.

These findings challenge the conventional view in the literature that predictability from firm characteristics diminishes as the cost of exploiting mispricing declines. Prior work such as Green, Hand, and Zhang (2017) documents a sharp drop in characteristics-based predictability in 2003, with only a small set of characteristics remaining as independent determinants. McLean and Pontiff (2016) also show that the returns to many characteristic-based anomalies fall with greater arbitrage activity or broader awareness of the anomalies. Our evidence indicates that the weak post-2000 performance of anomalies reflects time variation in return predictability. When we time portfolios to the temporal dynamics of predictability, the power of these characteristics largely reappears. The pricing-on strategy that embeds this timing delivers markedly superior performance to the pricing-off strategy.

C. *Ensembling “Weak” Signals*

So far, our analysis has focused on the portfolio level. In this section, we incorporate pricing-state identification into the combination of individual cross-sectional signals to predict individual stock returns. In Equation (1) of Section II, we forecast stock returns using cross-sectional univariate regressions for each predictor in an out-of-sample setting. Our sample uses 211 predictors, so at each time t , we have a pool of “weak” return forecasts. To improve the return forecast, we first consider a naïve approach that averages all forecasts. A simple average reduces the variance of forecast errors because individual biases and errors tend to cancel out. Specifically, the average combined signal is

$$Comb_{naïve}^{i,t+1} = \frac{1}{J_t} \sum_{j=1}^{J_t} \hat{r}_{j,i,t+1}, \quad (1)$$

where J_t represents the number of predictors at month t , and $\hat{r}_{i,j,t+1}$ is the return forecast for stock i based on signal j . Note that $Comb_{naïve}$ treats all individual forecasts equally.

We next incorporate pricing-state identification into the combination scheme:

$$Comb_{on}^{i,t+1} = \frac{1}{N_{on,t}} \sum_{j=1}^{J_t} \hat{r}_{j,i,t+1} \cdot I_{j,t}, \quad (2)$$

$$Comb_{off}^{i,t+1} = \frac{1}{N_{off,t}} \sum_{j=1}^{J_t} \hat{r}_{j,i,t+1} \cdot (1 - I_{j,t}), \quad (3)$$

where $I_{j,t}$ equals 1 when predictor j will be in the pricing-on state next month and 0 otherwise. $N_{on,t} = \sum_{j=1}^{J_t} I_{j,t}$ is the number of predictors identified as pricing-on for next month based on information at time t , and $N_{off,t}$ is the number of pricing-off predictors at time t . Thus, $Comb_{on}$ ($Comb_{off}$) aggregates the information from pricing-on (pricing-off) predictors. If pricing-state identification is informative about the predictive power of signals, we expect that $Comb_{on}$ will filter out noninformative predictors and therefore improve predictability relative to the simple average combination $Comb_{naïve}$.

We evaluate the economic value of the three forecast combinations by sorting stocks into deciles each month based on the signals available at that time and then forming a value-weighted long-short portfolio that buys the top decile and shorts the bottom decile. Panel A in Table [IA.II](#) reports average monthly returns for each decile and the high-minus-low spread with Newey-West t -statistics. In the full sample, $Comb_{on}$ delivers a strong and monotonic cross-sectional pattern that runs from -0.32 percent in decile 1 to 1.66 percent in decile 10. The high-minus-low spread is 1.98 percent with a t -statistic of 9.00, which is far larger than the 1.36 percent spread for $Comb_{naïve}$ with a t -statistic of 6.29. As expected, $Comb_{off}$ shows no cross-sectional power. Its spread is -0.12 percent and statistically insignificant. The same ordering holds in both subperiods. Before 2000, $Comb_{on}$ produces a spread of 2.38 percent with a t -statistic of 9.08, compared with 1.74 percent for $Comb_{naïve}$ and an insignificant 0.06 percent for $Comb_{off}$. After 2000, $Comb_{on}$ continues to generate a sizable 1.53 percent spread with a t -statistic of 4.19, whereas $Comb_{naïve}$ drops to 0.93 percent with a t -statistic of 2.40 and $Comb_{off}$ turns negative at -0.34 percent.

Panel B summarizes the performance of the corresponding value-weighted long-short portfolios. Over the full sample, $Comb_{on}$ achieves the highest annualized mean return of

Table IA.II

Ensembling Multivariate Weak Signals

This table compares the predictive power of three combined signals. $Comb_{naïve}$ is a simple averaged signal constructed using Equation (1). $Comb_{on}$ and $Comb_{off}$ are constructed using Equation (2), which takes into account signals' pricing state. At the end of each month, stocks are sorted into deciles by each forecast, value-weighted decile returns are computed for the next month, and the high-minus-low spread is reported with returns in percent per month and Newey-West t statistics in Panel A. Results are shown for the full sample from January 1975 to December 2023 and for two subsamples before and after 2000. Panel B reports performance statistics for the corresponding value-weighted long-short portfolios, including annualized mean return, volatility, skewness, Sharpe ratio, maximum drawdown, Calmar ratio, and factor-model adjusted returns using FF3, FF5, and HXZ with t -statistics.

Panel A: Portfolio Returns

	Low	2	3	4	5	6	7	8	9	High	H-L	t -stats
<i>Panel A.1: Full Sample, Jan. 1975 - Dec. 2023</i>												
$Comb_{naïve}$	0.08	0.58	0.66	0.82	0.89	1.01	1.04	1.16	1.28	1.44	1.36	(6.29)
$Comb_{on}$	-0.32	0.27	0.47	0.61	0.82	0.98	1.14	1.23	1.42	1.66	1.98	(9.00)
$Comb_{off}$	0.91	0.86	0.86	0.82	0.82	0.87	0.78	0.87	0.86	0.78	-0.12	(-0.49)
<i>Panel A.2: First Half, Jan. 1975 - Dec. 1999</i>												
$Comb_{naïve}$	0.04	0.62	0.77	0.95	0.98	1.18	1.20	1.36	1.53	1.78	1.74	(8.52)
$Comb_{on}$	-0.36	0.24	0.54	0.68	0.92	1.12	1.32	1.39	1.68	2.02	2.38	(9.08)
$Comb_{off}$	0.90	0.83	0.81	0.77	0.81	0.90	0.88	0.92	0.95	0.96	0.06	(0.21)
<i>Panel A.3: Second Half, Jan. 2000 - Dec. 2023</i>												
$Comb_{naïve}$	0.12	0.53	0.54	0.67	0.79	0.81	0.87	0.94	0.99	1.05	0.93	(2.40)
$Comb_{on}$	-0.28	0.32	0.39	0.53	0.70	0.82	0.94	1.06	1.13	1.25	1.53	(4.19)
$Comb_{off}$	0.92	0.90	0.92	0.87	0.84	0.83	0.67	0.83	0.75	0.58	-0.34	(-0.83)

Panel B: Long-Short Portfolio Performance

	Ann. $\bar{R}$	Ann.Vol	Skew	SR	MDD	Calmar	FF3	t -stats	FF5	t -stats	HXZ	t -stats
<i>Panel B.1: Full Sample, Jan. 1975 - Dec. 2023</i>												
$Comb_{naïve}$	16.33	16.67	0.15	0.98	53.46	0.31	1.42	(8.48)	0.97	(5.68)	0.72	(4.37)
$Comb_{on}$	23.76	19.54	-0.10	1.22	40.76	0.58	2.13	(9.79)	1.79	(6.53)	1.45	(5.53)
$Comb_{off}$	-1.49	18.69	-1.34	-0.08	93.69	-0.02	-0.16	(-0.73)	-0.51	(-2.36)	-0.54	(-1.78)
<i>Panel B.2: First Half, Jan. 1975 - Dec. 1999</i>												
$Comb_{naïve}$	20.90	12.84	-0.37	1.63	16.57	1.26	1.58	(7.79)	1.35	(6.92)	1.09	(4.74)
$Comb_{on}$	28.56	17.20	0.52	1.66	29.57	0.97	2.36	(7.10)	2.26	(6.86)	1.86	(4.31)
$Comb_{off}$	0.76	15.76	-2.49	0.05	77.63	0.01	-0.23	(-0.71)	-0.54	(-1.71)	-0.51	(-1.15)
<i>Panel B.3: Second Half, Jan. 2000 - Dec. 2023</i>												
$Comb_{naïve}$	11.16	20.04	0.43	0.56	53.46	0.21	1.30	(5.98)	0.59	(3.82)	0.55	(2.32)
$Comb_{on}$	18.33	21.79	-0.36	0.84	40.76	0.45	1.95	(7.26)	1.35	(5.09)	1.31	(4.97)
$Comb_{off}$	-4.04	21.50	-0.72	-0.19	91.30	-0.04	-0.19	(-0.57)	-0.60	(-1.90)	-0.66	(-1.49)

23.76 percent and the highest Sharpe ratio of 1.22, compared with 16.33 percent and 0.98 for $Comb_{naïve}$ and -1.49 percent and -0.08 for $Comb_{off}$. $Comb_{on}$ also reduces downside risk. Its maximum drawdown is 40.76 percent versus 53.46 percent for $Comb_{naïve}$ and 93.69

percent for $Comb_{off}$, and its Calmar ratio improves to 0.58. Risk-adjusted results reinforce these conclusions. The FF3, FF5, and HXZ alphas for $Comb_{on}$ are large and highly significant, while $Comb_{off}$ alphas are negative and insignificant and $Comb_{naive}$ alphas are positive but smaller.

The subperiod evidence highlights the robustness of $Comb_{on}$. Before 2000, $Comb_{on}$ delivers an annualized mean of 28.56 percent and a Sharpe ratio of 1.66, both above $Comb_{naive}$ at 20.90 percent and 1.63. $Comb_{off}$ remains weak with a Sharpe ratio of 0.05 and a large drawdown of 77.63 percent. After 2000, $Comb_{on}$ continues to outperform with an annualized mean of 18.33 percent and a Sharpe ratio of 0.84, while $Comb_{naive}$ falls to 11.16 percent and 0.56 and $Comb_{off}$ deteriorates to -4.04 percent and -0.19. $Comb_{on}$ maintains materially smaller drawdowns than the alternatives in this period and retains strong and significant factor-model alphas.

Overall, the decile patterns and portfolio statistics show that incorporating pricing-state identification into the combination of signals markedly strengthens the cross-sectional forecast and its economic value. $Comb_{on}$ consistently dominates $Comb_{naive}$, while $Comb_{off}$ carries little or no predictive content, especially in the post-2000 period.

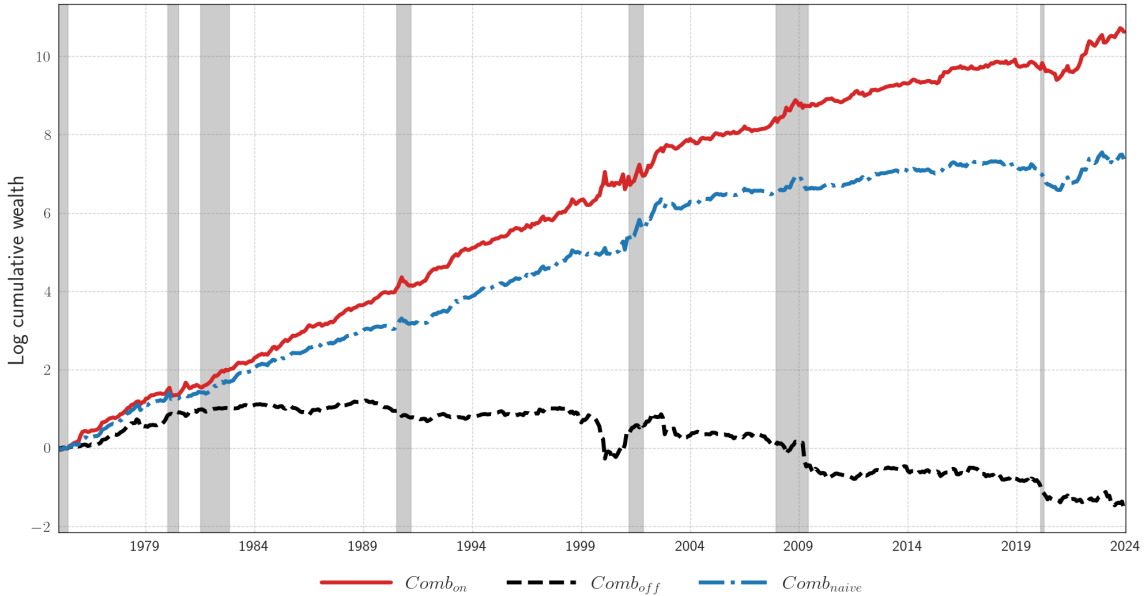


Figure IA.6. Cumulative wealth of combination portfolios. This figure plots cumulative wealth for value-weighted long-short portfolios constructed from three combined forecasts, $Comb_{on}$, $Comb_{off}$, and $Comb_{naive}$. At each month end, stocks are sorted into deciles using each forecast, the portfolio buys the top decile and shorts the bottom decile, and the portfolio is rebalanced monthly. Wealth starts at one dollar and is displayed on a log scale over the sample from January 1975 to December 2023. Shaded bands mark NBER recessions.

Figure IA.6 plots log cumulative wealth from an initial one dollar for the three combination strategies. $Comb_{on}$ compounds steadily and remains above the other two lines throughout the sample, with particularly strong gains in the late 1990s, the mid-2000s, and again after 2020. The shaded recession bands coincide with only brief pauses for $Comb_{on}$, and the line resumes its upward trajectory quickly, which is consistent with the smaller maximum drawdowns and higher Sharpe ratios reported in the table. $Comb_{naïve}$ delivers slower compounding. It keeps pace with $Comb_{on}$ until the mid 1980s, then trails increasingly, flattens after 2010, and shows a modest recovery only near the end of the sample. $Comb_{off}$ fails to accumulate wealth. It suffers a sharp break around the dot-com bust, another decline around the global financial crisis, and a persistent drift downward thereafter.

The figure visually confirms that incorporating pricing-state identification into the signal combination produces material economic value. $Comb_{on}$ filters out low-quality forecasts and concentrates exposure in the set of strong predictors, which yields smoother compounding, quicker recoveries around recessions, and large long-run wealth gains. $Comb_{naïve}$ is also informative but lacks timing and therefore underperforms, while $Comb_{off}$ contains almost non-predictive content.